# PRISMATIC PRESENTATIONS: THE GEOMETRIC COUNTERPART OF TRICKLE GROUPS

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Abstract. We introduce the concept of prismatic presentations, a family of presentations defined geometrically that includes trickle groups (right-angled Artin groups, right-angled Coxeter groups, cactus groups, and more) and twisted right-angled Artin groups. We develop geometric tools that allow us to prove that groups having prismatic presentations enjoy the (shortlex) falsification by fellow-traveller property and are (shortlex) automatic.

Contents



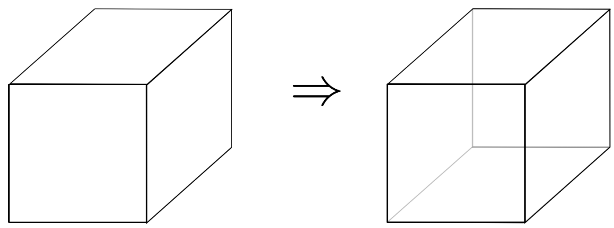
## 1. Introduction

In this work we introduce a type of presentations whose associated disc diagrams exhibit a good geometric behaviour. This provides tools to exploit the geometry of interesting families of groups, among which *trickle groups* [3] must be highlighted, as their study from the geometric point of view has been the main motivation of our work. The formal definition of the presentations that we work with has a similar flavour to those in small cancellation theory but, as it happens in that context, the apparently complicated definitions have a simple geometric interpretation that, in our case, motivates the name: *prismatic presentations.*

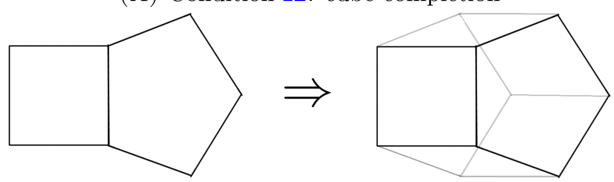
Informally, we can say that a presentation $\langle \Sigma | R \rangle$ is prismatic if the set of relators admits a partition $R = S \sqcup H$ such that, in the Cayley complex $X$ associated to $\langle \Sigma | R \rangle$, the following properties hold:

- **I1** (Squares and heads) The 2-cells in $X$ associated to $S$-relators are all squares, whereas the $H$-relator 2-cells can be polygons of 2 or more sides.
- **I2** (Cube completion) If $X$ contains three $S$-relator squares glued as in the corner of a cube, then the whole cube must be present in $X$.
- **I3** (Prism completion) If an $S$-relator 2-cell and a $H$-relator 2-cell share exactly one edge, then $X$ contains a prism where one of the bases is the $H$-relator,

one of the lateral faces is the $S$-relator and they are glued along their overlapping side.



(A) Condition **I2**: cube completion



(B) Condition **I3**: prism completion

FIG. 1. Properties of a prismatic presentation

The formal definition of prismatic presentations includes two more conditions regarding the overlap between the different relators (see Section 3), but conditions **I2** and **I3** already illustrate the key point of these presentations: any disc diagram over a prismatic presentation admits *hexagonal* and *prismatic moves* (see Fig. 2). This means that, whenever we see a hexagon tessellated by 3 squares in a disc diagram, we can substitute this subdiagram by the alternative tessellation (given by the other side of the cube, which we know exists by the cube completion condition). Similarly, if we see a subdiagram which can be seen as part of a full prism, we can retessellate it by its complement in the prism (using the prism completion condition). Hexagonal moves appear in the literature in the context of non positively curved cube complexes (see [20], for example) and they will also play an important role in our proofs. As for prismatic moves, we introduce them to deal

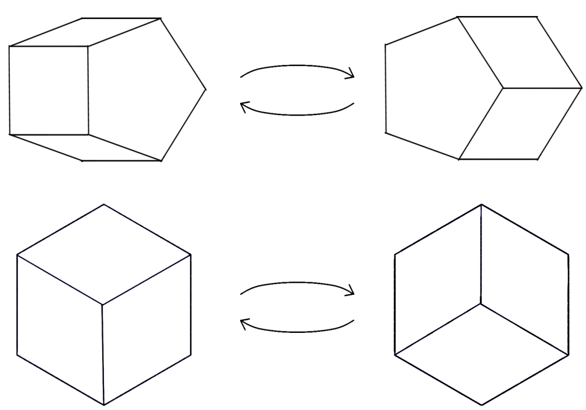


FIG. 2. Prismatic moves (above) and hexagonal moves (below)

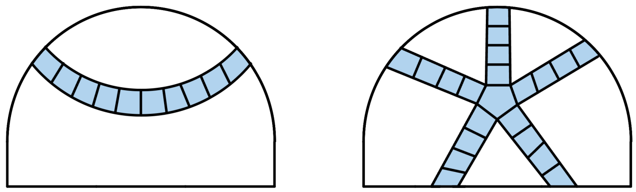


FIG. 3. Corridor (left) and spider (right).

with torsion relations in trickle groups and, in the more general context of prismatic presentations, their main purpose is to "move around" the $H$-relator 2-cells in a disc diagram, so they will also be heavily used in our proofs.

Due to the nature of the definition, it not clear that building concrete examples of prismatic presentations is easy. However, in Section 7.1 we prove the following:

**Theorem A** (Theorem 7.11)**.** *The standard presentation of a trickle group is prismatic.*

As a consequence we obtain many examples of groups with a prismatic presentation: the family of trickle groups, introduced in [3], includes cactus groups, right-angled Artin groups, right-angled Coxeter groups and, in general, graph products of cyclic groups. In addition to trickle groups, in Section 7.2, we see that twisted right-angled Artin groups (TRAAGs) and a generalisation of these admit prismatic presentations. It follows from [2] and [9] that twisted right-angled Artin groups are not trickle groups (Remark 7.12), so it is clear that our results can be applied to a large class of groups.

**LAM:** he quitado lo del WP original, habría que modificarlo si este es el sitio donde queremos meterlo. Merece la pena pensar a futuro, qué es lo que pasa cuando miramos sílabas (subwords de una cabeza?)

Disc diagrams over prismatic presentations can be seen as the union of two types of subdiagrams, formally defined in Section 4: *corridors*, which are bands of squares parallel to a given edge, and *spiders*, which are given by an $n$-gonal head together with the $n$ corridors induced by its $n$ sides (see Fig. 3). The geometry of prismatic presentations is exploited through the following result, which characterises geodesics in these presentations in terms of corridors and spiders.

**Theorem B** (Theorem 5.21)**.** *Let $\langle \Sigma \mid S \sqcup H \rangle$ be a prismatic presentation, and let $w$, $u$ be words over its generators representing the same group element. Then, $w$ is geodesic if and only if no disc diagram $D$ for $wu^{-1}$ that is free of* pathologies *and* siamese spiders *contains:*

*(1) A corridor whose both ends lie on $w \cap \partial D$, or*
*(2) A spider with the majority of its legs ending on $w \cap \partial D$.*

A disc diagram over a prismatic presentation is free of *pathologies* and *siamese spiders* if its corridors and spiders are well-behaved, e.g. by not exhibiting double intersections (Definitions 4.11 and 4.12). Corollary 4.28 shows that, given any word $w$ representing the identity over a prismatic presentation, there exists a disc diagram $D$ with $\partial D = w$ that is free of pathologies and siamese spiders.

From this characterisation of geodesics we recover a Tits-style rewriting system, described in Definition 5.25, similar to the one proposed for trickle groups in [3, §2.3]. The characterisation of geodesics also gives an important tool to understand the Cayley graph of a prismatic presentations. We say that a graph $\Gamma$ satisfies the *falsification by fellow-traveller property* (FFTP) if there exists a global constant $K$

**LAM:** Pongo lo de Tits aquí por ahora. Quería hacer referencia a la pregunta de BGP de la regularidad de las formas normales, pero he visto que esa pregunta es para las formas normales de su §4, no de §5. A primera vista parecen distintas.

such that for every non-geodesic path in $\Gamma$, there exists a shorter path with the same endpoints that $K$-fellow travels the non-geodesic path (that is, it remains uniformly at distance $K$ from the non-geodesic path ), and we say that a group $G$ with finite generating set $\Sigma$ has the FFTP if the Cayley graph of $G$ with respect to $\Sigma$ satisfies this property. The characterisation of geodesics, together with hexagonal and prismatic moves, allows us to show:

**Theorem C** (Theorem 5.24). *If a group $G$ with finite generating set $\Sigma$ admits a prismatic presentation with $\Sigma$ as the generating set, then the pair $(G, \Sigma)$ satisfies the falsification by fellow traveller property (FFTP).*

This geometric property has been shown to hold for many families of groups (see [14, 17, 5, 12, 13, 1]) and it has interesting implications for the groups that have it. Indeed, groups with FFTP are of type $F_3$, their Dehn function is at most quadratic (see [6]), they have a regular set of geodesics (see [15]) and, as a consequence, they have a rational growth function.

**LAM:** He comentado lo de la preg del rational growth, que Yago mencionó pero que no encontramos en el paper de BGP.

[**?**] states that if a group $\langle\Sigma\rangle$ has the FFTP, then its shortlex representatives form a regular language for any total ordering of $\Sigma$. This fact allows us to obtain another relevant result for prismatic presentations:

**Theorem D** (Theorem 6.4). *Every group admitting a finite prismatic presentation is* shortlex automatic*.*

This theorem partially answers a question posed in [3] regarding the automaticity of trickle groups. *Automaticity* is a desirable property for groups, as it allows one to study them from an algorithmic point of view. There is some previous work in the literature showing that some subfamilies of prismatic presentations are automatic (see [19, 11] for RAAGs and [8] for TRAAGs; in [9], the author shows that trickle groups are cocompactly cubulable and automaticity can be deduced from this). In our case, by proving that groups with a prismatic presentation are *shortlex* automatic, we provide a very specific automatic structure for this larger class of groups.

A group $G = \langle\Sigma\rangle$ is defined to be *shortlex automatic* if the set of minimal shortlex representatives of elements of $G$ is a regular language with respect to some total ordering of $\Sigma \cup \Sigma^{-1}$ and, for some constant $K$, any two words $w, v$ of this regular language that represent adjacent elements in the Cayley graph $K$-fellow travel. It is worth noting that the automatic structure that our result provides is not necessarily a biautomatic structure (see Example 6.5). Another family that is known to be shortlex automatic is that of Artin groups of large type (see [13]).

**LAM:** a mi no me pega mucho dar este ejemplo de large type, o al menos me parece que está desconectado... no sé cómo conectarlo, lo dejo.

The paper is structured as follows: in Section 2 we provide preliminary definitions and results (FFTP, trickle groups, TRAAGs, disc diagrams) and in Section 3 we define prismatic presentations. Section 4 contains the basic notions (corridors, spiders, pathologies) and results exploring the geometry of prismatic presentations, leading to a result that allows us to work with disc diagrams that are free of pathologies in the following sections. In Section 5 we use the previous tools to characterise geodesics and to show that groups with a finite prismatic presentation satisfy the FFTP. The tools from this section are also used in Section 6 as part of the proof of shortlex automaticity. Lastly, Section 7 shows that several families of groups (trickle, TRAAGs and a generalisation of them) admit prismatic presentations. The authors would like to observe that the definition of trickle groups in [3] allows for the existence of hexagonal moves, as it is possible to gather from Section 7 in the cited article.

## 2. Definitions and notation

2.1. **The Falsification by Fellow Traveller property.** Given a graph $X$, a *(combinatorial) path* $p$ is a sequence $v_0e_1v_1\dots e_nv_n$ where $v_0,\dots,v_n$ are vertices in $X$ and, for $i=1,\dots,n$, $e_i$ is an edge between $v_i$ and $v_{i+1}$. The *length* of $p$ is the number $n$, and the vertices $v_0$, $v_n$ are called the *endpoints* of the path. The path $p$ is *geodesic* if its length is minimal among the lengths of all the paths in $X$ with endpoints $v_0$,$v_n$.

**Definition 2.1** (Fellow traveller for graphs)**.** For $K\geq 0$, we say that two paths $p=v_1\dots v_n$, $q=u_1\dots u_m$ in a graph $X$ *asynchronously $K$-fellow travel* if there is a non-decreasing, proper map $f\colon \mathbb{N}\to\mathbb{N}$ satisfying that the vertices $v_i,u_{f(i)}$ are at distance at most $K$ in the graph for all $i\geq 0$. We say that $p$ and $q$ *synchronously $K$-fellow travel* if $v_i$, $u_i$ are at distance at most $K$ for all $i\geq 0$.

**Definition 2.2** (FFTP for graphs)**.** A graph $X$ has the *(asynchronous) falsification by fellow traveller property* if there is a constant $K$ such that for every non-geodesic path $p$ there exists a path $q$ with the same endpoints as $p$, that is shorter than $p$ and that $K$-fellow travels with it.

These concepts are translated to a group using its Cayley graph with respect to a given set of generators. For any group $G$ with finite generating set $S$, its elements are represented by words $w\in(S\cup S^{-1})^*$. Each such word $w$ corresponds to a combinatorial path in the Cayley graph of $G$ with respect to $S$ starting at a fixed basepoint, take the basepoint to be the vertex corresponding to the identity $1_G$.

The length of such a word $w$, which coincides with the length of the corresponding path, will be denoted by $|w|$. The length of the shortest $v\in(S\cup S^{-1})^*$ such that $v=_G w$ is denoted by $|w|_G$. We say $w$ is geodesic if its corresponding path is geodesic or, equivalently, if $|w|=|w|_G$.

**Definition 2.3** (Fellow traveller for groups)**.** Given a group $G$ with finite generating set $S$ and a constant $K\geq 0$, we say that two words $w,v\in(S\cup S^{-1})^*$ *asynchronously $K$-fellow travel* if the corresponding paths $p$, $q$ asynchronously $K$-fellow travel in the Cayley graph of $G$ with respect to $S$. We say that $w$ and $v$ *synchronously $K$-fellow travel* if $p$ and $q$ synchronously fellow travel in the Cayley graph.

The falsification by fellow traveller property for groups was introduced by Neumann and Shapiro in [16]. Two notions appear in the literature: the synchronous and the asynchronous falsification by fellow traveller property. In [6], Elder showed that these two notions are actually equivalent, so we choose to work with the asynchronous one.

**Definition 2.4** (FFTP for groups)**.** A group $G$ with generating set $S$ is said to have the *falsification by fellow traveller property* (FFTP) *with respect to* $S$ if the Cayley graph of $G$ with respect to $S$ has the asynchronous falsification by fellow traveller property.

Similarly, we can define a *shortlex* version of FFTP associated to the shortlex ordering.

**LAM:** He metido esto aquí aunque sea provisionalmente para no tener que definirlo en la Section 6. He cambiado el alfabeto $X$ por $S$, $X$ aquí es un grafo.

**Definition 2.5** (Shortlex order)**.** Let $(S,<)$ be a totally ordered alphabet. We will consider two different order relations in $S^*$:

- The *lexicographical order*: $w_1, w_2 \in S^*$, $w_1 <_{\text{lex}} w_2$ if and only if $w_1$ is a strict prefix of $w_2$ or the letter in the first position in which $w_1$ and $w_2$ differ is smaller for $w_1$.
- The *shortlex order*: for $w_1, w_2 \in S^*$, $w_1 <_{\text{SL}} w_2$ if and only if $w_1$ is shorter than $w_2$ or they have the same length and $w_1 <_{\text{lex}} w_2$.

**Notation 2.6** $(\mathrm{SL}(g), \mathrm{SL}_G)$**.** The shortlex order is a well-ordering. In particular, if $G$ is a group generated by $\Sigma$ and we consider the shortlex order on $(\Sigma\cup\Sigma^{-1})^*$ with respect to some ordering of $\Sigma\cup\Sigma^{-1}$, there is a well-defined minimal representative for each element $g$ of the group $G$ under this order, which will be denoted $\mathrm{SL}_{(\Sigma,<)}(g)$ or simply $\mathrm{SL}(g)$ if $(\Sigma,<)$ is understood by the context. We will denote by $\mathrm{SL}_{(G,\Sigma,<)}$ the language of these minimal representatives of elements of $G$, and write $\mathrm{SL}_G$ for short if $(G,\Sigma,<)$ is understood.

Now, we can define the shortlex version of the falsification by fellow traveller property:

**LAM:** esto se aleja un poco del paralelismo con la definiciones anteriores (def grafo- def grupo)... pero no pega escribir SL-fftp para grafos. lo dejo así, que opinen paloma y yago

**Definition 2.7** (SL-FFTP for groups)**.** Let $G$ be a group with generating set $\Sigma$, and fix $<$ a total ordering on $\Sigma$. The group $G$ is said to have the *shortlex falsification by fellow traveller property* (SL-FFTP) *with respect to* $(\Sigma,<)$ if there exists a constant $K \geq 0$ such that, for any $w \notin \mathrm{SL}_G$, there exists a word $v \in (\Sigma\cup\Sigma^{-1})^*$ that $K$-fellow-travels with $w$ and satisfies that $w =_G v$ and $v <_{\text{SL}} w$.

We recall these two statements, taken from [**?**]:

**Lemma 2.8.** *If $G$ satisfies* FFTP *with respect to $\Sigma$, then $G$ satisfies* SL-FFTP *with respect to $\Sigma$.*

For the next statement, recall that a *language* over an alphabet $S$ is a set of finite words over the alphabet, and that a language $L$ over an alphabet $S$ is *regular* if it is the language accepted by a *finite state automaton* over $S$. See [7, Chapters 1.1 and 1.2] for the precise definitions.

**Proposition 2.9.** *Let $G$ be a group with finite generating set $\Sigma$ satisfying* FFTP*, then* $\mathrm{SL}_G$ *is a regular language.*

2.2. **Trickle groups.** In this section, we follow closely [3] in order to introduce trickle groups, making only small changes in notation that are useful for our purposes.

Let $\Gamma$ be a simplicial graph. The set of vertices of $\Gamma$ is denoted by $V(\Gamma)$ and its set of edges is denoted by $E(\Gamma)$. The *star* of a vertex $x \in V(\Gamma)$, denoted by $\mathrm{Star}_\Gamma(x)$, is the full subgraph of $\Gamma$ spanned by $\{y \in V(\Gamma) : \{x,y\} \in E(\Gamma)\} \cup \{x\}$. We will simply write $\mathrm{Star}(x)$ when $\Gamma$ can be clearly determined from the context.

**Definition 2.10.** A *trickle graph* is a quadruple $(\Gamma, \leq, \mu, (\varphi_x)_{x\in V(\Gamma)})$, where

- $\Gamma$ is a simplicial graph,
- $\leq$ is a (partial) order on $V(\Gamma)$,
- $\mu$ is a labelling $\mu\colon V(\Gamma) \to \mathbb{N}_{\geq 2} \cup \{\infty\}$ of the vertices,
- $\varphi_x\colon \mathrm{Star}_\Gamma(x) \to \mathrm{Star}_\Gamma(x)$ is a graph automorphism of $\mathrm{Star}_\Gamma(x)$ for all $x \in V(\Gamma)$.

Let $x, y, z \in V(\Gamma)$. To keep the notation light we write $y_x$ instead of $\varphi_x(y)$; with this convention, $(\varphi_x \circ \varphi_y)(z)$ becomes $(z_y)_x$, which we simplify to $z_{yx}$. In this way, the expression $z_{xy_x}$ represents $(\varphi_{y'} \circ \varphi_x)(z)$, where $y' = \varphi_x(y)$. Moreover, we will write $y_{x^a}$ to refer to $\varphi_x^a(y)$ for $a \in \mathbb{Z}$. As usual, the notation $x < y$ means that $x \leq y$ and $x \neq y$. For $x, y \in V(\Gamma)$, the notation $x||y$ means that $x$ and $y$ are not comparable in the sense that $x \not\leq y$ and $y \not\leq x$. We set $E_{||}(\Gamma) = \{\{x, y\} \in E(\Gamma) : x||y\}$.
The quadruple $(\Gamma, \leq, \mu, (\varphi_x)_{x \in V(\Gamma)})$ must satisfy the following conditions:

(a) For all $x, y \in V(\Gamma)$, if $x < y$, then $\{x, y\} \in E(\Gamma)$.
(b) For all $x, y, z \in V(\Gamma)$, if $\{x, y\} \in E_{||}(\Gamma)$ and $z \leq y$, then $\{x, z\} \in E_{||}(\Gamma)$.
(c) For all $x \in V(\Gamma)$ and all $y, z \in \mathrm{Star}_\Gamma(x)$, we have $z \leq y$ if and only if $z_x \leq y_x$.
(d) For all $x \in V(\Gamma)$ and all $y \in \mathrm{Star}_\Gamma(x)$, if $y_x \neq y$, then $y < x$.
(e) For all $x \in V(\Gamma)$, if $\mu(x)$ is finite, then $\varphi_x$ has finite order and its order divides $\mu(x)$.
(f) For all $x \in V(\Gamma)$ and all $y \in \mathrm{Star}_\Gamma(x)$, $\mu(y_x) = \mu(y)$.
(g) For all $x, y, z \in V(\Gamma)$, if $z < y < x$, then $z_{yx} = z_{xy_x}$.

We will often say that $\Gamma$ is a trickle graph meaning that, implicitly, $\leq, \mu$ and $(\varphi_x)_{x \in V(\Gamma)}$ are also given.

**Definition 2.11.** The *trickle group* $\mathrm{Tr}(\Gamma)$ associated with a trickle graph $(\Gamma, \leq, \mu,$ $(\varphi_x)_{x \in V(\Gamma)})$ is the group defined by the following presentation:

$$\mathrm{Tr}(\Gamma) = \langle V(\Gamma) \mid x^{\mu(x)} = 1 \text{ for } x \in V(\Gamma) \text{ s.t. } \mu(x) \neq \infty, y_x x = x_y y \text{ for } \{x, y\} \in E(\Gamma) \rangle.$$

**Remark 2.12.** Condition (d) in the definition of a trickle graph ensures that all the relations in the presentation of $\mathrm{Tr}(\Gamma)$ are of the form $xy = zx$ and $x^\mu = 1$. Indeed, for $\{x, y\} \in E(\Gamma)$, we have that the relation $y_x x = x_y y$ becomes

- $yx = x_y y$ if $x < y$,
- $y_x x = xy$ if $y < x$, or
- $yx = xy$ if $x||y$.

When working with trickle groups, it is useful to keep in mind the example of *cactus groups* since, unlike graph products of cyclic groups, cactus groups can exhibit quasi-commutation relations $xy = y'x$ where $y \neq y'$.

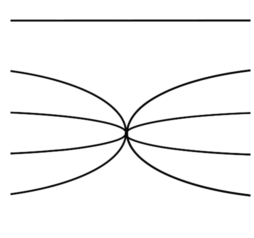
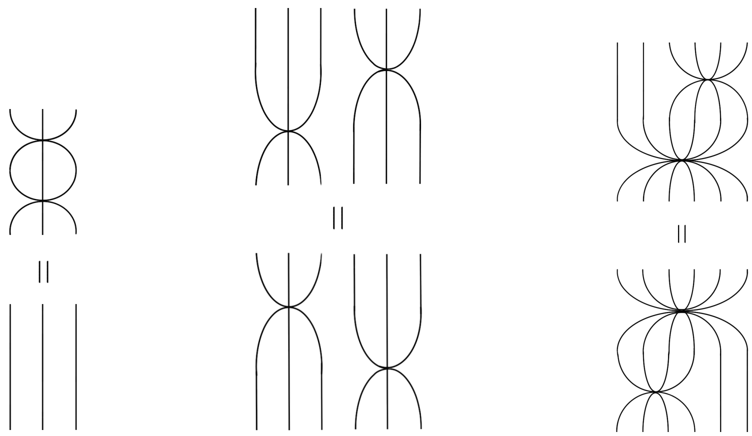


FIG. 4. A generator in a cactus group corresponding to an interval of size 4.

**Definition 2.13.** The cactus group $J_n$ for $n \in \mathbb{N}_{\geq 2}$ is given by a presentation with generators $x_I$ for each interval $I \subset [1, n]$ (Fig. 4) and relations $x_I^2 = 1$, $x_I x_J = x_J x_I$ if the subintervals $I$, $J$ are disjoint and $x_I x_J = x_{J'} x_I$ for intervals $[a, b] = J \subset I = [c, d]$, where the interval $J'$ defined as $[c+d-b, c+d-a]$ is the reflection of $J$ under the central symmetry of $I$ (Fig. 5). As shown in these figures, the elements in this group and the relations given in the previous presentation can be represented using braided-like pictures.

**LAM:** estaría bien alinear estas imágenes, pero no he sabido.



(A) The square of a generator is trivial. (B) Commuting generators. (C) Quasi commuting generators.

FIG. 5. Relations in a cactus group shown with braid-like diagrams.

2.3. **Twisted right-angled Artin groups.** Twisted right-angled Artin groups (T-RAAGs) are defined by a presentation, which is graphically encoded in what is known as a mixed graph.

**Definition 2.14** (Mixed graph)**.** A *mixed graph* $\Gamma = (V, E, D, \iota, \tau)$ consists of an *underlying simplicial graph*$(V, E)$, a set of *directed edges* $D \subseteq E$, and maps $\iota, \tau \colon D \to V$ giving the *initial* and *terminal* vertex of edges. For each $e = \{x, y\} \in D$, we have $\iota(e), \tau(e) \in \{x, y\}$ and $\iota(e) \neq \tau(e)$.

**Definition 2.15** (Decorated mixed graph)**.** A *decorated mixed graph* $\Gamma = (V, E, D, \iota, \tau, \mu)$ is a mixed graph $\Gamma = (V, E, D, \iota, \tau)$ equipped with a labelling map $\mu \colon V \to \mathbb{N} \cup \{\infty\}$.

**Notation 2.16.** Let $\Gamma$ be a mixed graph $(V, E, D, \iota, \tau)$ or a decorated mixed graph $(V, E, D, \iota, \tau, \mu)$.

(i) If $e = \{a, b\} \in E \setminus D$, we use the notation $[a, b]$, or equivalently $[b, a]$, to refer to the edge $e$.

**LAM:** He puesto esta frase un poco distinta, para que no me dé dolor de cabeza leer $e = [a, b]$, sino que sea "we use the notation...". Si quieres volver al original, descoméntalo.

(ii) If $e = \{a, b\} \in D$, we use the notation $[\iota(e), \tau(e)\rangle$ to refer to the edge $e$.

Graphically, edges $[a, b]$, and $[a, b\rangle$ are respectively depicted as follows:


A mixed graph determines a T-RAAG as follows:

**Definition 2.17** (Twisted RAAG)**.** Let $\Gamma = (V, E)$ be a mixed graph. Define

$$T(\Gamma) = \big\langle V \mid ab = ba \text{ if } [a,b] \in E - D, \quad aba = b \text{ if } [a,b\rangle \in D \big\rangle.$$

We call $T(\Gamma)$ the *twisted right-angled Artin group* based on $\Gamma$, and $\Gamma$ its *defining graph.*

Similarly, we can define a torsion T-RAAG:

**Definition 2.18** (torsion T-RAAG)**.** Let $\Gamma = (V, E, D, \iota, \tau, \mu)$ be a decorated mixed graph. The *torsion twisted right-angled Artin group* based on $\Gamma$ is given by the quotient

$$T_{tor}(\Gamma) = T(\Gamma)/\langle\langle a^{\mu(a)} = 1 \text{ for } a \in V \text{ with } \mu(a) < \infty \rangle\rangle,$$

where $T(\Gamma)$ denotes the twisted RAAG based on the underlying mixed graph of $\Gamma$.

2.4. **Disc diagrams.** In order to work with trickle groups, we make use of a geometric tool that helps us understand words representing the identity element of the group. Essentially, we will have diagrams (see the definitions below) that encode how a word representing the trivial element can be seen as a product of conjugates of relators in the given presentation of the group. We present these ideas in the style of [10], giving the necessary background to understand how we use these diagrams.

**Definition 2.19.** A map $f\colon X \to Y$ between 2-dimensional CW-complexes is *combinatorial* if it maps open cells homeomorphically to open cells. A complex is combinatorial if all attaching maps are combinatorial.

**Definition 2.20** (Disc diagram)**.** A *disc diagram* $D$ is a compact contractible combinatorial 2-complex, together with an embedding $D \hookrightarrow \mathbb{E}^2$. Let $X$ be a 2-complex. A *disc diagram in* $X$ is a disc diagram $D \hookrightarrow \mathbb{E}^2$ together with combinatorial map $D \to X$. The *area* of a disc diagram $D$ is the number of 2-cells in $D$.

In our setup, we will work with disc diagrams in a specific complex:

**Definition 2.21** (Presentation complex)**.** Given a group presentation $P = \langle S \mid R \rangle$, its *standard presentation complex* has a single 0-cell $v$, a 1-cell attached (and labelled) by each generator in the presentation, and a polygonal 2-cell for each relator in the presentation which is attached along the corresponding 1-cells that label the relator. The *Cayley complex* $X_P$ of the presentation is the universal cover of the standard presentation complex.

**Remark 2.22.** In the context of a CW-complex, we will use the term *path* to refer to a combinatorial paths in their 1-skeleton as described in Section 2.1. The 1-cells of a disc diagram $D$ in the Cayley complex $X_P$ inherit a natural labelling from the one in $X_P$, and so a path in $D$ carries a label determined by the labels of its edges.

**Definition 2.23.** A *boundary cycle* of a disc diagram $D$ in $X_P$ is a closed path that travels along the whole boundary of the disc diagram respecting the planar embedding of the diagram. A *boundary label* of $D$ is a word on the generators of the presentation $P$ which labels a boundary cycle of $D$.

Notice that the map from a disc diagram $D$ to the Cayley complex $X_P$ is not necessarily an immersion:

**Definition 2.24.** A disc diagram $D$ in $X_P$ is *reduced* if it does not contain two 2-cells $B_1$, $B_2$ sharing a 1-cell $e$ in $\partial B_1 \cap \partial B_2$ in a way that the label of the boundary of both cells starting at the shared one-cell coincides (that is, $\partial B_1$ and $\partial B_2$ map to the same closed path in $X_P$).

**Remark 2.25.** It is possible to replace a non reduced disc diagram $D$ in $X_P$ with another disc diagram $D'$ in $X_P$ which has the same boundary cycle but smaller area: just remove $e$ and the interior of $B_1$ and $B_2$ and then glue $\partial B_1 - e$ with $\partial B_2 - e$.

Since any word labelling a closed path in the 1-skeleton of the Cayley complex represents the identity element of the group given by the presentation, it is clear that the boundary label of any disc diagram in a presentation complex represents the identity. A converse statement is also true (see [4] for a proof).

**Lemma 2.26** (Van Kampen's Lemma)**.** *Given a presentation for a group and a freely reduced word $w$ over the generators that represents the identity element, there is a (reduced) disc diagram in the corresponding Cayley complex whose boundary cycle is labelled by $w$.*

**Notation 2.27.** Let $D$ be a disc diagram with boundary label $w$ and let $u$ be a subword of $w$. We will write $\partial_u D$ to mean $\partial D \cap u$, respectively, where $u$ can be seen as a 1-subcomplex in the boundary of $D$.

**LAM:** Yago, esta definicion que usamos es muy intuitiva pero nos estamos haciendo un lio con escribirla formalmente. Escribimos algo, vemos que tiene un caso raro y pasamos a reescribir una definición más complicada. Si tienes sugerencias, dilas xfavor.

**LAM:** la def de enclosed by puede hacer cosas raras si las curvas $\gamma$ hacen cosas raras, pero creo que no nos importa porque nosotras siempre lo usamos con curvas normalitas.

**Definition 2.28.** Let $D$ be a disc diagram and let $\gamma_1, \gamma_2, \ldots, \gamma_n$ be 1-dimensional subcomplexes of $D$ whose union is connected. The diagram $D$ embeds in the plane, so we can consider the space $\mathbb{R}^2 - \cup_{i=1}^n \gamma_i$ which is a disjoint union of simply connected components (the *interior* components) and a connected component that is not simply connected (the *exterior* component). Let $I$ be the subcomplex of $D$ given by the union of the interior components with $\cup_{i=1}^n \gamma_i$. We will refer to the minimal connected subcomplex of $I$ containing all the interior components as the *region enclosed by* $\gamma_1, \gamma_2, \ldots, \gamma_n$.

## 3. Prismatic presentations

In this section we introduce our main object of study: prismatic presentations. The definition of this family of group presentations, despite its combinatorial appearance, carries an intuitive geometric interpretation which allows us to infer powerful properties of the groups they define. We first share a more intuitive point of view which should be kept in mind when reading the formal definitions that can be found below.

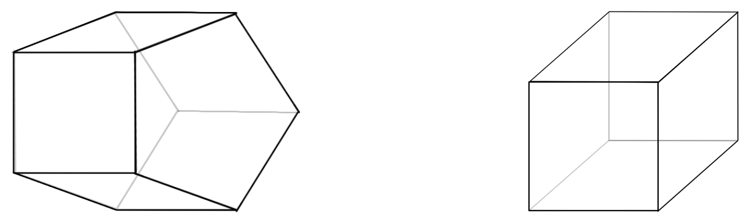
A prismatic presentation is one where the set of relations can be split into two families: *squares*, which are relations of length four, and relations of varying length giving polygons which we called *heads*. These relations must satisfy some requirements: for example, one should not be able to glue a square to another 2-cell along a subword of length more than one (*small pieces* condition) and any two heads that can be glued along at least one edge should define another head relation (*fusion of heads* condition). The conditions imposed on the set of relations give rise to two natural objects, *cubes* and *prisms*, which lead to further constraints in the definition of prismatic presentation. These constraints ensure that one can perform two operations on a disc diagram over a prismatic presentation: *hexagonal moves* and *prismatic moves*, which can be visualized in Fig. 8.

Let us formally define the basic objects that give geometric meaning to the definition of prismatic presentation that we will present afterwards.

**Definition 3.1.** A *prism* with base of length $n$ is a $CW$-complex $P$ with the following structure:

- The 0-skeleton of $P$ consists of $2n$ vertices.
- The 1-skeleton of $P$ is built by attaching $3n$ 1-cells to the 0-skeleton as we describe next. Choose $V_1 \sqcup V_2$ a partition of the vertices of the 0-skeleton and enumerate them so that we have $V_1 = \{v_1, \dots, v_n\}$ and $V_2 = \{v_{n+1}, \dots, v_{2n}\}$. Attach $2n$ 1-cells joining vertices $v_i$ and $v_{i+1}$ for $i \in \{1, \dots, n-1\} \cup \{n+1, \dots, 2n-1\}$. Attach the remaining $n$ 1-cells so that they join $v_i$ and $v_{i+n}$ for $i \in \{1, \dots, n\}$.
- For every cycle of length four in the 1-skeleton that contains two vertices of $V_1$ and two vertices of $V_2$, attach a square 2-cell, which will be referred to as a *lateral face*. For the cycles of length $n$ with sets of vertices $V_1$ and $V_2$ attach respective 2-cells, which must be the interior of a polygon of $n$ sides and will be called the *bases* of the prism. As a result, the 2-skeleton of $P$ consists of $(2+n)$ 2-cells and $P$ is simply connected.

See Fig. 6(A)



(A) Prism with pentagonal base. (B) Cube.

FIG. 6. Prisms and cubes.

**Definition 3.2.** A *cube* is a prism with base of length 4 (see Fig. 6(B)).

**Definition 3.3.** Let $\Sigma$ be an alphabet. A $\Sigma$-prism ($\Sigma$-cube) is a prism (cube) whose 1-cells are labelled by elements of $\Sigma \cup \Sigma^{-1}$.

**Remark 3.4.** Given a $\Sigma$-cube with three pairwise adjacent 2-cells $\sigma_1, \sigma_2, \sigma_3$ with the labelling indicated on the left, we can always label the remaining 2-cells of the $\Sigma$-cube $\tau_1, \tau_2, \tau_3$ as indicated on the right:

$$\begin{aligned} \sigma_1 &= x_1 a_1 b_1 y_1^{-1}, & \tau_1 &= x_2 b_2 a_3 y_2^{-1}, \\ \sigma_2 &= y_1 a_2 b_2 z_1^{-1}, & \tau_2 &= y_2 b_3 a_1 z_2^{-1}, \\ \sigma_3 &= z_1 a_3 b_3 x_1^{-1}, & \tau_3 &= z_2 b_1 a_2 x_2^{-1}. \end{aligned}$$

Notice that boundary of the pairwise adjacent 2-cells $\sigma_1, \sigma_2, \sigma_3$ and $\tau_1, \tau_2, \tau_3$ respectively is an hexagon with the same labelling in both cases. See Fig. 8.

**Remark 3.5.** Given a more general $\Sigma$-prism, we can use the following standard notation for its labelling: the boundaries of the bases $\tau$ and $\rho$ are labelled by

$$\tau = l_1 l_2 \dots l_n \quad \text{and} \quad \rho = r_1 r_2 \dots r_n,$$

and the lateral faces $\sigma_1, \dots, \sigma_n$ will have their boundaries labelled by

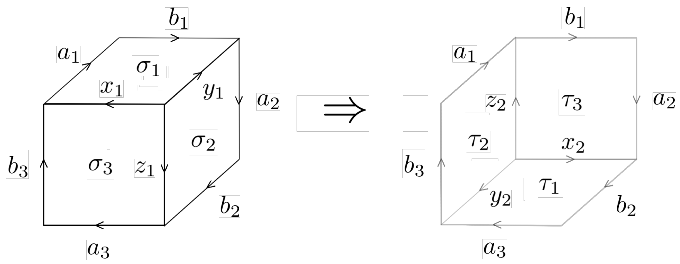
$$\sigma_i = l_i m_i^{-1} r_i^{-1} m_{i-1} \quad \text{for} \quad i = 1, \dots, n,$$

under the convention $m_0 = m_n$. See Fig. 8.

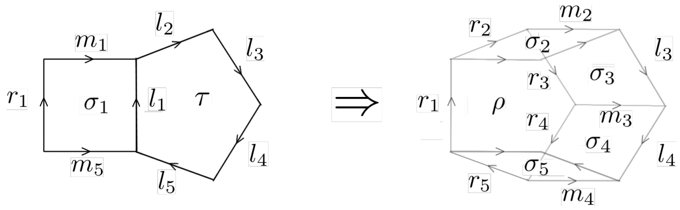
**LAM:** label escogido para que el combinatorial map de trickle funcione si mandamos $l_i \mapsto x$, $r_i \mapsto x_y$, $m_i \mapsto y_{x^{m-i}}$. ASEGURARSE

We now may give the definition of the type of presentation we will be working with for the rest of the paper. The reader should note that the notation in this definition has been chosen so as to match the one in Remarks 3.4 and 3.5. .

**LAM:** en estas imágenes no estamos muy contentas con la claridad de las etiquetas. He añadido una mini descripción en la caption de qué son 2-cells y qué 1-cells, a pesar de que eso ya está definido en **P3**,**P4**. Qued muy repetitivo, o facilita la lectura completar la caption así?




(A) Condition **P3**: cube completion. $\sigma_i, \tau_i$ denote 2-cells, $a_i, b_i, x_i, y_i, z_i$ denote 1-cells.




(B) Condition **P4**: prism completion. $\sigma_i, \tau, \rho$ denote 2-cells, $m_i, l_i, r_i$ denote 1-cells.

FIG. 7. Conditions for prismatic presentation following notation in Definition 3.6, see Fig. 1 for an unlabelled version.

**Definition 3.6** (Prismatic presentation)**.** Let $\langle \Sigma | R \rangle$ be a group presentation with generating set $\Sigma$ and a set of cyclically reduced relations $R$ for a group $G$. We say that this presentation is prismatic if the set of relations admits a partition $R = S \sqcup H$ satisfying the following conditions:

- **P1** (Squares and heads) Every $\sigma \in S$ is a word of length 4. The cells in the corresponding Cayley complex associated to these relations will be called squares, and cells associated to $H$ will be called heads.
- **P2** (Small pieces) Given $\sigma \in S$ and $\tau \in S \sqcup H$, no cyclic permutation of $\sigma$ or $\sigma^{-1}$ shares a subword of length 2 with $\tau$.
- **P3** (Cube completion) For every triple $\sigma_1, \sigma_2, \sigma_3$ of cyclic permutations of relations in $S \cup S^{-1}$ that can be factorized as

$$\sigma_1 = x_1 a_1 b_1 y_1^{-1}$$
$$\sigma_2 = y_1 a_2 b_2 z_1^{-1}$$
$$\sigma_3 = z_1 a_3 b_3 x_1^{-1},$$

where $a_i, b_i, c_i, x_1, y_1, z_1 \in \Sigma \cup \Sigma^{-1}$ and $a_1 b_1 a_2 b_2 a_3 b_3$ is a cyclically reduced word, there exists another triple $\tau_1, \tau_2, \tau_3$ of cyclic permutations of relations $S \cup S^{-1}$ such that

$$\tau_1 = x_2 b_2 a_3 y_2^{-1}$$
$$\tau_2 = y_2 b_3 a_1 z_2^{-1}$$
$$\tau_3 = z_2 b_1 a_2 x_2^{-1}.$$

Note that these six relations provide the labelling of a $\Sigma$-cube. See Fig. 7(A).

**P4** (Prism completion) For every $\sigma_1$ that is a cyclic permutation of a relation in $S \cup S^{-1}$ and $\tau$ a cyclic permutation of a relation in $H \cup H^{-1}$, where

$$\sigma_1 = l_1 m_1^{-1} r_1^{-1} m_n,$$
$$\tau = l_1 l_2 \dots l_n,$$

with $l_i, m_1, r_1 \in \Sigma \cup \Sigma^{-1}$, there is a relation $\rho$ that is cyclic permutation of an element in $H \cup H^{-1}$ and there are relations $\sigma_2, \dots \sigma_n$ that are a cyclic permutation of elements in $S \cup S^{-1}$ of the form

$$\rho = r_1 r_2 \dots r_n,$$
$$\sigma_i = l_i m_i^{-1} r_i^{-1} m_{i-1},$$

with $r_i, l_i, m_i \in \Sigma \cup \Sigma^{-1}$. Note that these $(n+2)$ relations provide the labelling of a $\Sigma$-prism. See Fig. 7(B).

**P5** (Fusion of heads) If $\sigma$ and $\tau$ are cyclic permutations of relations in $H \cup H^{-1}$ of the form $\sigma = uv$ and $\tau = v^{-1}w$ with $v \neq \epsilon$, then the cyclic reduction of $uw$ is a cyclic permutation of a relation in $H \cup H^{-1}$ or the empty word.

Note that this definition does not guarantee the existence of cubes and prisms *in the Cayley complex*, but the existence of combinatorial maps from cubes and prisms into it. However, the fact that these cubes and prisms embed injectively into the Cayley complex will be a consequence of the solution to the word problem obtained in Theorem 5.21.

This definition naturally guarantees closure under taking *graph products*. Consider $\Gamma = (V\Gamma, E\Gamma)$ a simplicial graph and a collection $\{G_v = \langle \Sigma_v \mid S_v \cup H_v \rangle\}$ of prismatic presentations associated to the vertices in $V\Gamma$. Consider the generating set $\Sigma = \sqcup_{v \in V\Gamma} \Sigma_v$, the set of heads $H = \sqcup_{v \in V\Gamma} H_v$ and the set of squares

$$S = (\sqcup_{v \in V\Gamma} S_v) \cup \big\{ ab = ba \ \text{ for } \ a \in \Sigma_u,\ b \in \Sigma_w,\ \{u, w\} \in E\Gamma \big\}.$$

The presentation $\langle \Sigma \mid H \cup R \rangle$ defines de graph product $\mathcal{G} = \mathcal{G}(\Gamma, \{G_v\}_{v \in \Gamma})$ of the groups $G_v$ associated to the graph $\Gamma$.

**Proposition 3.7.** *Let $\langle \Sigma \mid H \cup R \rangle$ be the presentation, as described above, of the graph product $\mathcal{G}(\Gamma, \{G_v\}_{v \in \Gamma})$ coming from a simplicial graph $\Gamma = (V\Gamma, E\Gamma)$ and a collection of prismatic presentations $\{G_v = \langle \Sigma_v \mid S_v \cup H_v \rangle\}$. The presentation $\langle \Sigma \mid H \cup R \rangle$ is a prismatic presentation.*

**LAM:** Esto habrá que ver si se quita, se mueve o qué. A ver si Yago nos da una opinión cuando lea todo.

*Proof.* Take $H$ to be the set of heads and $S$ to be the set of squares. Because adjacent letters in the new squares (the ones coming from generators $a \in \Sigma_u,\ b \in \Sigma_w$ where $\{u, w\} \in E\Gamma$) come from disjoint generating sets $\Sigma_u, \Sigma_w$, these squares do not share pieces of length two with other heads or squares. This, together with the fact that $\langle \Sigma_v \mid S_v \cup H_v \rangle$ is a prismatic presentation for $v \in V\Gamma$, ensures that **P2** is satisfied. Condition **P5** is trivially satisfied because all the heads in $H$ come from heads in $H_v$ for some vertex $v \in V\Gamma$ and heads coming from different vertices do not share any edges.

To see that conditions **P3** and **P4** are satisfied, first note that the new squares involving generators from different generating sets are always commuting generators, meaning that opposite edges of these squares are labelled by the same generating (and have the same direction).

For **P3**, consider squares $\sigma_1, \sigma_2, \sigma_3$ as in Definition 3.6. There are three possible cases. First, if at least two squares belong to the same collection $S_v$ for some

$v \in V\Gamma$, the third one does as well and the cube completion condition is satisfied because of $\langle \Sigma_v \mid S_v \cup H_v \rangle$ being prismatic. Second, assume without loss of generality that $\sigma_1 \in H_u$ for $u \in V\Gamma$ and $\sigma_2$ is a commutation square coming from some $\{u, w\} \in E\Gamma$. This means that $\sigma_3$ is also a commutation square coming from the edge $\{u, w\} \in E\Gamma$, and the square is completed by taking $\tau_1$ to be (a cyclic permutation of) $\sigma_1$ and $\tau_2$, $\tau_3$ to be the necessary commutation squares between a generator in $\Sigma_u$ and a generator in $\Sigma_w$. For the last case, if all $\sigma_1$, $\sigma_2$ and $\sigma_3$ are commutation squares coming from the edges in $E\Gamma$, the cube completion is satisfied by taking $\tau_i$ to be equal to (a cyclic permutation of) $\sigma_i$ for $i = 1, 2, 3$.

**LAM:** Respondiendo a la pregunta que paloma escribió en el correo: todos los $\sigma_i$ son iguales sólo si $\tau$ es la torsión de algún generaror.

For **P4**, consider a head $\tau$ and a square $\sigma_1$ as in Definition 3.6. If both relations belong to the same prismatic presentation $\langle \Sigma_v \mid S_v \cup H_v \rangle$, the prism can be completed. Assume that $\tau \in H_u$ for some $u \in V\Gamma$ and $\sigma_1$ is a commutation square coming from $\{u, w\} \in E\Gamma$. Similar to before, because parallel sides in commutation squares have the same direction and label, the head $\rho$ can be taken to be equal to $\tau$ and the squares $\sigma_2, \ldots, \sigma_n$ will be the necessary commutation squares between generators in $\Sigma_u$ and generators in $\Sigma_w$. □

From the geometric point of view, conditions **P3** and **P4** in the definition above guarantee that we can make the following substitutions when working with disc diagrams over a prismatic presentation:

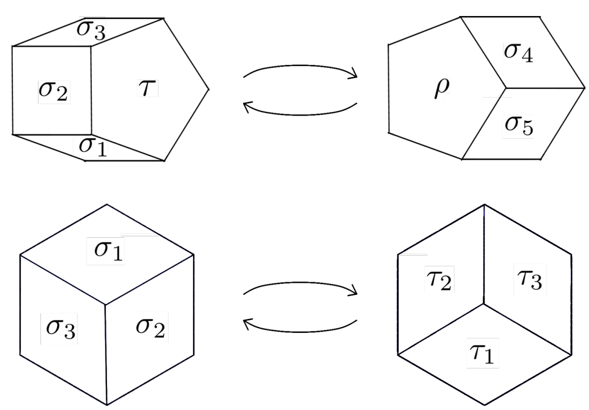


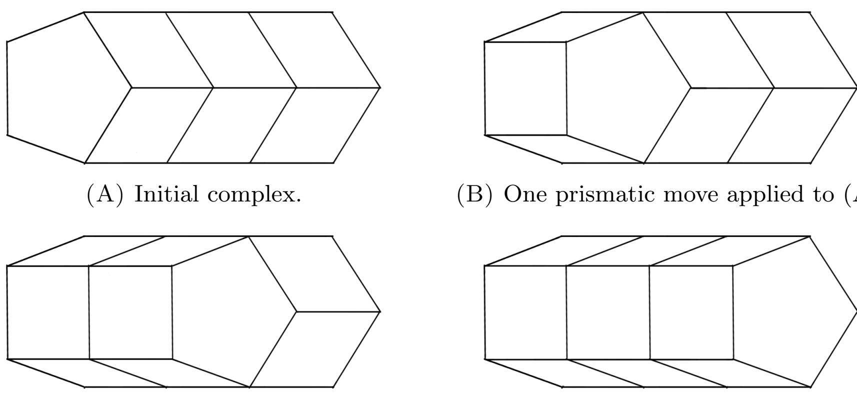

Fig. 8. Prismatic moves (above) and hexagonal moves (below).

**Definition 3.8.** (Half-cube and hexagonal move) Consider a disc diagram $D$ over a prismatic presentation $\langle \Sigma | S \sqcup H \rangle$. A half-cube (or tessellated hexagon) in $D$ is a subdiagram of $D$ induced by three pairwise adjacent 2-cells that are associated to relations $\sigma_1, \sigma_2, \sigma_3 \in S$ as in **P3**. Condition **P3** ensures that, whenever $D$ contains a half-cube induced by relations $\sigma_1, \sigma_2, \sigma_3$ as a subdiagram, this subdiagram can be retessellated using the other half of the cube that $\sigma_1, \sigma_2$ and $\sigma_3$ induce, given by the corresponding relations $\tau_1, \tau_2, \tau_3$. This retessellation is what we call a *hexagonal move* (see Fig. 8).

**Definition 3.9.** (Half-prism and prismatic move) Consider a disc diagram $D$ over a prismatic presentation $\langle \Sigma | S \sqcup H \rangle$. A half-prism in $D$ is a subdiagram consisting of one 2-cell associated to a relation $\tau \in H$ and several 2-cells associated to relations $\sigma_1, \sigma_{i+1}, \ldots, \sigma_{i+k}$ as in **P4**. Condition **P4** ensures that, whenever $D$ contains a

$n$-gonal half-prism induced by relations $\tau \in H$ and $\sigma_i, \ldots, \sigma_{i+k} \in S$ for some $k \geq 1$ as a subdiagram, this subdiagram can be retessellated using the remaining 2-cells of the prism that the half-prism induces, given by the corresponding relations $\rho \in H$ and $\sigma_{k+1}, \ldots, \sigma_n \in S$. This retessellation is what we call a *prismatic move* (see Fig. 8).

**Remark 3.10.** If a disc diagram over a prismatic presentation contains a half-prism with one base, its retessellation contains the other base that has been 'pushed away' (see Fig. 8). This will be used to choose disc diagrams where the placement of the bases is convenient for our purposes.



(A) Initial complex.

(B) One prismatic move applied to (A).

(C) One prismatic move applied to (B).

(D) One prismatic move applied to (C).

Fig. 9. Prismatic moves used sequentially to deal with longer prisms, where first and last complex are understood as prisms of height three.

**Remark 3.11.** Prisms have been defined to have height equal to 1 (that is, the distance between the two bases is 1), but we will often apply prismatic moves sequentially to work with disc diagrams that contain 'prisms' of height greater than 1 as shown in Fig. 9. This is strictly a sequence of applications of prismatic moves but, for short, we will refer to this sequence simply as a prismatic move.

## 4. Corridors, spiders and pathologies

By condition **P1** in the definition of a prismatic presentation, we distinguish two types of 2-cells in the disc diagrams associated to such a presentation: squares and heads. When a diagram does not contain any heads, we talk about *squared diagrams*:

**Definition 4.1** (Squared diagram)**.** Let $\langle \Sigma | S \sqcup H \rangle$ be a prismatic presentation. We say that a disc diagram $D$ over this presentation is *squared* if all its 2-cells are squares (that is, they are associated to relations in $S$).

**Definition 4.2** (Corridor)**.** Let $D$ be a disc diagram over a prismatic presentation. We say that two edges $e$ and $e'$ of $D$ are *parallel* if they are opposite sides of a square 2-cell in $D$. The transitive closure of the relation that identifies parallel edges gives an equivalence relation defined on the set of edges of $D$. Given an edge

$e$ of $D$, the *corridor induced by* $e$ is the subcomplex induced by the 2-cells of $D$ which contain two parallel edges in this equivalence class.

**Definition 4.3** (Length of a corridor)**.** Consider a disc diagram $D$ and a corridor $C$ which is non-self-intersecting, the *length* of $C$ in $D$ is given by the number of 2-cells in the $C$. A zero-length corridor will also be referred to as a *trivial* corridor.

**Definition 4.4** (Spider)**.** Given a disc diagram $D$ over a prismatic presentation $\langle \Sigma | S \sqcup H \rangle$, a *head* is a polygonal 2-cell in $D$ arising from a relation in $H$. The (possibly trivial) corridors in $D$ induced by the edges of a head are called *legs*. The subdiagram of $D$ given by the union of a head and its associated legs is what we call a *spider* diagram.

When we choose disc diagrams over prismatic presentations, there are certain configurations that corridors and spiders can exhibit that we wish to avoid. We call them *pathological* configurations (Definition 4.11).

We begin with the four corridor configurations that we want to avoid. We will call them *corridor pathological configurations* and each of them is illustrated in Fig. 10. Notice that the corridors involved in these pathologies might be legs of spiders.

**Definition 4.5** (Circular corridor)**.** Let $C$ be a corridor. If there is a sequence of edges $(e_i)_{i=1}^n$, with $n \geq 2$, defining the corridor such that $e_i$ is parallel to $e_{i+1}$ and $e_1 = e_n$, we say that $C$ is a *circular* corridor.

Identifying each square in the corridor with the space $[-1,1] \times [-1,1]$, where the points $(0, \pm 1)$ correspond to the midpoints of the edges defining the corridor, we define the *fibre* of the circular corridor as the simple, closed curve given by the union of the segments $\{0\} \times [-1,1]$ for each square in the corridor.

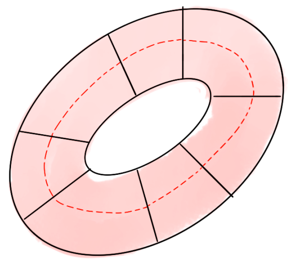
**Definition 4.6** (Self-osculating corridor)**.** Given a corridor $C$, if there is a sequence of edges $(e_i)_{i=1}^n$ defining the corridor such that $e_i$ is parallel to $e_{i+1}$ and $e_1$, $e_n$ share exactly one vertex, we say that the corridor $C$ *self-osculates*.

If the sequence $(e_i)_{i=1}^n$ is minimal, in the sense that there are no edges $e_i$, $e_j$ with $1 \leq i < j < n$ and $e_i$, $e_j$ sharing exactly one vertex, we define its fibre as follows. With the notation above, take the segments $\{0\} \times [-1,1]$ for the squares induced by the pairs $(e_i, e_{i+1})$. The union of these segments, together with the straight line joining the midpoint of $e_1$ with the midpoint of $e_n$, is a simple, closed curve which we call the *fibre* of the self-osculation.

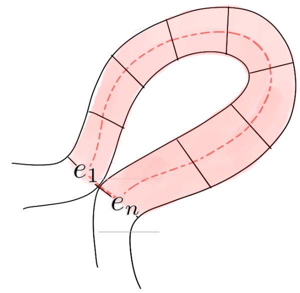
**Definition 4.7** (Self-intersecting corridor)**.** A corridor $C$ *self-intersects at a square* $s$ in $C$ if there is a sequence of edges $(e_i)_{i=1}^n$ defining the corridor such that $e_i$ is parallel to $e_{i+1}$ and $e_1$, $e_n$ are adjacent edges of the square $s$.

If the sequence is minimal, in the sense that there is no shorter subsequence exhibiting this self-intersecting behaviour, we consider again the segments $\{0\} \times [-1,1]$ for the squares induced by the pairs $(e_i, e_{i+1})$ in the sequence. The union of these segments with the straight line in $s$ that joins the midpoint of $e_1$ to the midpoint of $e_n$ is a simple, closed curve called the *fibre* of the self-intersection.

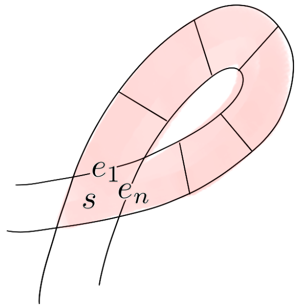
**Definition 4.8** (Doubly intersecting corridors)**.** Two distinct corridos $C_1$, $C_2$ intersect at a square $s$ in $C_1 \cap C_2$ if there are edges $e$, $f$ inducing $C_1$ and $C_2$ respectively that are adjacent sides of the square $s$. We say that there is a *double intersection* between two corridors if they intersect at two different squares. In this case, we



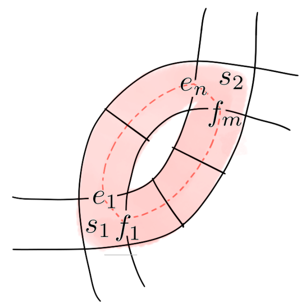
(A) Circular corridor




(B) Self-osculating corridor




(C) Self-intersecting corridor




(D) Doubly intersecting corridors

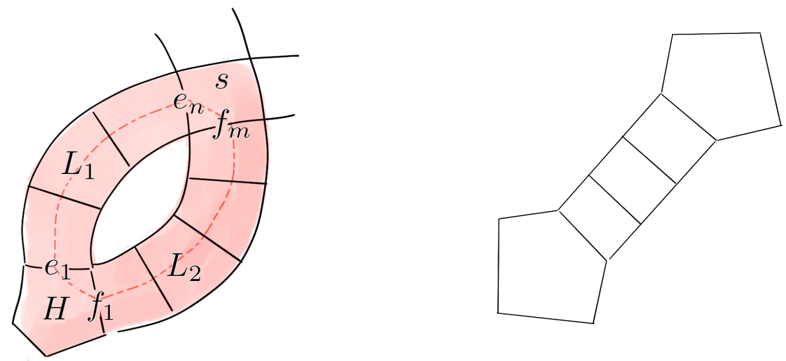
FIG. 10. In each image the dotted line shows a fibre and the shaded cells show the corresponding pathology $P$.

can define sequences $(e_i)_{i=1}^n$ and $(f_i)_{i=1}^m$ of edges inducing $C_1$ and $C_2$ respectively so that the pairs $(e_i, e_{i+1})$ and $(f_i, f_{i+1})$ give parallel edges, $e_1$, $f_1$ are consecutive edges in a square $s_1$ in $C_1 \cap C_2$, $e_n$, $f_m$ are consecutive edges in a square $s_2$ in $C_1 \cap C_2$.

If these sequences are minimal in the sense that no subsequences exhibit a double intersection, we define its fibre as follows. We take once more the segments $\{0\} \times [-1, 1]$ for the squares induced by the pairs $(e_i, e_{i+1})$ and by the pairs $(f_i, f_{i+1})$, and we take the union of these segments with the straight line in $s_1$ joining the midpoints of $e_1$ and $e_n$, and the straight line in $s_2$ joining the midpoints of $e_n$ and $f_m$. This union yields what we call the *fibre* of the double intersection, which is again a simple, closed curve.

We also have pathologies involving a head, and we call them *spider pathological configurations*. They can be of two types:

**Definition 4.9** (Intersecting legs)**.** Given a spider, we say that two of its legs $L_1$ and $L_2$ intersect at a square $s \in L_1 \cap L_2$ if there are edges $e$, $f$ inducing $L_1$ and $L_2$ respectively that are adjacent sides of the square $s$. If the sequences of parallel edges $\{e_i\}^n \subset L_1$ and $\{f_i\}^m \subset L_2$, where $e_1$, $f_1$ are edges in the head and $e_n$, $f_m$ are consecutive sides of $s$, are chosen to be minimal in the sense that no pair $(e_i, f_j) \neq (e_n, f_m)$ exhibits an intersection of $L_1$ and $L_2$, we define the *fibre* of the leg intersection as follows. As we did with doubly intersecting corridors, we take the segments $\{0\} \times [-1, 1]$ for all squares with opposite sides $(e_i, e_{i+1})$ and $(f_i, f_{i+1})$. Taking the union of these segments together with a straight line in $s$ joining the




(A) Two intersecting legs (B) Siamese spiders sharing one leg.

FIG. 11

midpoint of $e_n$ and $f_m$ and a straight line in the head $H$ joining the midpoint of $e_1$ to the midpoint of $f_1$, we obtain a simple, closed curve which we call fibre of the intersection.

**Definition 4.10** (Circular leg)**.** Suppose that $e$ and $f$ are two edges in the boundary of a head that induce the same corridor $L$. Then we say that $L$ is a circular leg.

Identifying each square in the leg $L$ with the space $[-1, 1] \times [-1, 1]$, where the points $(0, \pm 1)$ correspond to the midpoints of the edges defining the corridor, we define the *fibre* of the circular leg as the simple, closed curve given by the union of the segments $\{0\} \times [-1, 1]$ for each square in the leg together with a segment in the head joining the midpoints of $e$ and $f$.

**Definition 4.11** (Pathology, internal area)**.** In the phenomena defined above, the 2-complex spanned by the 2-cells whose interiors intersect the fibre is what we call a *pathology* $P$. We will say that $P$ is a corridor pathology if it comes from a corridor pathological configuration and we will say that $P$ is a spider pathology if it comes from a spider pathological configuration. Given a pathology $P$ in a disc diagram $D$, Jordan's theorem ensures that its fibre separates the diagram in an interior and an exterior region. The 2-complex given by the union of the interior region with the pathology $P$ is what we call the *diagram enclosed by the pathology*, denoted $\bar{P}$. The *internal area* of the pathology $a(P)$ is given by the number of 2-cells in $\bar{P} - P$.

There is one last configuration of cells, this time involving two spider heads, that will be important later on:

**Definition 4.12** (Siamese spiders)**.** If two spiders $S_1$, $S_2$ with heads $H_1$, $H_2$ share a leg $L$, we say that $S_1$ and $S_2$ are siamese through the leg $L$.

**LAM:** Creo que no nos importa, pero esto crea patologías a no ser que cojamos un conjunto maximal de patas que $H_1$ y $H_2$ comparten. PD incluso si coges cjto maximal patas, puedes crear un corredor circular si comparten dos patas no contiguas. No sé si deberíamos hacer un apunte al respecto.

**Lemma 4.13.** *Let $D$ be a disc diagram over a prismatic presentation. If $D$ contains siamese spiders, there is an alternative disc diagram $D'$ with $\partial D = \partial D'$ that contains no siamese spiders.*

*Proof.* We check that each pair of heads $H_1$, $H_2$ sharing a leg $L$ can be removed by a retessellation of the diagram. We first perform a (sequence of) prismatic moves along $H_1 \cup L$ to "push" $H_1$ towards $H_2$. Formally, $H_1 \cup L$ gets retessellated to the union of a head $H_1'$ and a set of legs, where $H_1'$ and $H_2$ share an edge. We distinguish two cases now: if $H_1'$ and $H_2$ are induced by the same relation and the diagram $H_1' \cup H_2$ is not reduced (that is, $\partial H_1'$ and $\partial H_2$ label the same word if read starting at their common edge), we can retessellate $H_1' \cup H_2$ to a diagram with area zero given by an edge path. Otherwise, if $H_1' \cup H_2$ yield a reduced diagram, Condition **P5** (fusion of heads) in the definition of prismatic presentations ensures that $H_1' \cup H_2$ can be retessellated by a single head relation, reducing the number of siamese spiders in the original diagram $D$. By repeated application we obtain the desired diagram $D'$. □

We gather in the following definitions some important concepts that will appear in the theorems of this section.

**Definition 4.14** (Side of a corridor)**.** Let $D$ be a squared disc diagram. Let $C$ be the corridor induced by an edge $e$ in $D$. Since $D$ can be embedded on the plane, it cannot contain a non-orientable surface. Therefore, if we remove the (interior of the) 1-cells of $C$ that are in the equivalence class of $C$ and all of the 2-cells, we obtain two one-dimensional complexes that we will call the *sides of the corridor*.

**Definition 4.15** (Arc)**.** Let $D$ be a disc diagram over a prismatic presentation. An *arc* in $D$ is a non-pathological corridor whose equivalence class contains two different edges $e$ and $e'$ of $\partial D$. The edges $e$ and $e'$ are called the *extremes* of the arc.

**Definition 4.16** (Outermost arc, outermost boundary arc)**.** Let $D$ be a squared disc diagram over a prismatic presentation. We say that an arc $A$ with extremes $e$ and $e'$ is *outermost* with respect to one of the components of $\partial D - \{e, e'\}$ if all the corridors induced by the edges in this component intersect the arc $A$. We say that an arc $A$ with extremes $e$ and $e'$ is an *outermost boundary arc* if one of its sides is contained in $\partial D$ and it is outermost with respect to both components of $\partial D - \{e, e'\}$.

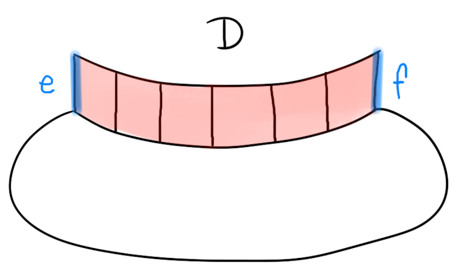



FIG. 12. Diagram with an arc at its boundary.

The rest of the section focuses on showing the following two results:

(1) Every squared diagram $D$ over a prismatic presentation admits a sequence of hexagonal moves and *collapses* (Definition 4.22) that leads to an equivalent squared diagram $D'$ without pathologies satisfying that two edges in $\partial D = \partial D'$ induce a corridor if and only if the corresponding edges in $\partial D'$ induce a corridor. Theorem 4.26.

(2) For every diagram $D$ over a prismatic presentation, there exists another diagram $D'$ which is free of pathologies and siamese spiders and satisfies that $\partial D = \partial D'$ (Corollary 4.28).

where we show that, if we have a corridor $B$ running along the boundary of a pathology-free squared disc diagram $D$ (what we call an *arc*, Definition 4.15) and this corridor is minimal in the sense that all other corridors in $D$ begin and/or end in the arc, we can iteratively use hexagonal moves to push the 2-cells in $D - B$ through $B$ (see Fig. 13).

The following is an auxiliary lemma that will be used in Theorem 4.20.

**Lemma 4.17.** *Let $D$ be a squared disc diagram over a prismatic presentation. Assume $D$ contains no pathologies. If $D'$ is a disc diagram obtained from $D$ by applying a hexagonal move, then $D'$ contains no pathologies.*

*Proof.* Let $D'$ be the resulting disc diagram after applying a hexagonal move to $D$. We will see that any pathology in $D'$ must come from a pathology in $D$. Let $H$ be a tesselated hexagon in $D$ and let $H'$ be its alternative tesselation in $D'$. Since $\partial H = \partial H'$ and $D \setminus H$ remains the same after retessellating, for any edge $e$ of $\partial H \cup (D - H)$ in $D$, there is a natural corresponding edge $e'$ of $\partial H' \cup (D' - H')$ in $D'$. We may think of the corridor $C_{e'}$ (induced by $e'$) as being the corridor $C_e$ (induced by $e$) after the retessellation. Notice that only the corridors induced by edges in $\partial H$ might change after the hexagonal move. Also, observe that two opposite edges in $\partial H'$ induce the same corridor. Let $e$ be an edge of $\partial H$ and suppose that $C_{e'}$ is involved in some pathology.

- If $C_{e'}$ is a circular corridor so must be $C_e$, since these two corridors are determined by the same edges in $\partial H = \partial H'$ and $D$, $D'$ only differ in the hexagon.
- If $C_{e'}$ is a self-intersecting corridor, and the self-intersection point occurs out of $H'$, the self-intersection occurs in $D$ for the same reason as before: $D$ and $D'$ only differ in the hexagon and the pairs of edges in $\partial H = \partial H'$ determining corridors have not changed. If the self-intersection point occurs in $H'$ it is immediate to see that there was already a self-intersecting point in $H$ for $C_e$.
- If $C_{e'}$ is a self-osculating corridor and the self-osculating point occurs outside of $H'$, it is immediate to see that the self-osculation was already exhibited by $C_e$ in $H$. If the self-osculating point appears in $H'$, note that it cannot be the interior vertex of $H'$ because the self-osculation vertex must belong, by definition, to at least 4 squares. Then it must be one of the vertices in $\partial H'$ and the two squares of $H$ adjacent to this point are part of $C_{e'}$. If we undo the hexagonal move, we now have a square of $H$ adjacent to the vertex corresponding to the self-osculating point in $D'$. Take the edge of $\partial H = \partial H'$ that lies inside the interior region of the pathology in $D'$. The corridor induced by this edge crosses $C_e$ twice: inside and outside $H$.
- If $C_{e'}$ exhibits a double intersection with none of the two intersection squares belonging to $H'$, it is clear from the fact that the hexagonal move only affects $H'$ that the same two corridors exhibited a double intersection with $C_e$ in $D$. If one of the intersection squares belongs to $H'$, call $C_{f'}$ the other corridor involved in the intersection and call $C_f$ the corresponding

corridor in $D$. Undoing the hexagonal move on $H'$, it is also clear that one of the squares in $H$ is an intersection square for two corridors defined by $C_e$ and $C_f$, so the double intersection was exhibited by $D$.

□

**Definition 4.18** (Perpendicular corridors)**.** Let $D$ be a disc diagram over a prismatic presentation. Two corridors $C$ and $C'$ are perpendicular at a square $s$ in $D$ if there exist edges $e$ and $e'$ in their respective equivalence classes that are adjacent sides of the square $s$.

**Notation 4.19.** To clarify the terminology in the next theorem, we note that when we say that two squares in a disc diagram are adjacent, we mean that they share at least one of their sides. In particular, two squares that share only one vertex are not adjacent.

**Theorem 4.20.** *Let $D = B \sqcup R$ be a squared disc diagram over a prismatic presentation. Assume that $D$ is free of pathologies, $B$ is an outermost boundary arc and $R$ has positive area. Then there are adjacent squares $t$ and $t'$ in $B$ and a square $s$ in $R$ adjacent to both of them. In particular, one can perform a hexagonal move that yields a new diagram $D' \sqcup s'$ where:*

*(1) $D' = B' \sqcup R'$ where $B'$ is an outermost boundary arc of $D'$,*
*(2) The area of $R'$ is smaller than the area of $R$,*
*(3) $s'$ is a square 2-cell which does not intersect $R'$,*
*(4) $\partial D = \partial(D' \sqcup s')$,*
*(5) The arcs $B$ and $B'$ are induced by the same two edges of $\partial D$ , and*
*(6) $D' \sqcup s'$ has no pathologies.*

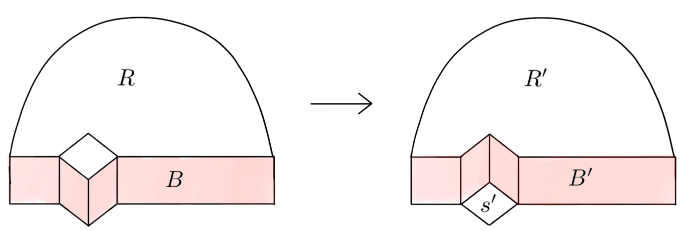



FIG. 13. Setup of Theorem 4.20

*Proof.* Since $B$ is an outermost boundary arc and $D$ contains no pathologies, every corridor in $D$ crosses $B$ exactly once. This, together with the fact that $a(R) > 0$, implies that there exists a square $s$ of $R$ that is adjacent to some square of $B$.

If $s$ is adjacent to two non-adjacent squares $t$ and $t'$ of the arc, then $D$ contains a pathology, which is a contradiction. Indeed, suppose $e$ is the edge shared by $s$ and $t$ and $e'$ is the edge shared by $s$ and $t'$. If $e$ and $e'$ are opposite edges of $s$, there is a double crossing between $B$ and the corridor perpendicular to $B$ at $t$. If $e$ and $e'$ are adjacent edges in $s$, then $B$ self-osculates.

Therefore, there are only two cases:

- There is a square of $R$ which is adjacent to two adjacent squares of $B$, or
- every square of $R$ adjacent to $B$ shares at most one edge with one of the squares of $B$.

Since self-osculation is forbidden, the first case yields a tesselated hexagon that admits a hexagonal move that pushes $B$ against $R$, reducing by 1 the area of $R$. Formally, $D$ is retesselated to $D' \sqcup s'$, with $s'$ one of the squares in the new tesselation of the hexagon to which we have performed the hexagonal move. Notice that (1) is satisfied. Indeed, the edges of the hexagon induce three corridors before and after the retessellation. In $D$, the two corridors that intersect the hexagon which are not $B$ intersect $B$. In $D'$, the two corridors induced by the same edges also intersect $B'$, so $B'$ is outermost with respect to the component of the boundary which does not intersect $s'$. It is clear that the side of $B'$ that intersects $s'$ is contained in $\partial D'$, so $B'$ is an outermost boundary arc in $D'$. Now, as $s'$ and $R'$ intersect different sides of $B'$, it is clear that (2) and (3) hold in the new diagram. Since the boundary of the hexagon remains invariant, property (4) holds and there is still an arc $B'$ induced by the same edges as $B$ (satisfying property (5)). Notice that, due to Lemma 4.17, the new diagram still has no pathologies, so property (6) is satisfied. To conclude, we only need to rule out the second case.

For the second case, we assume that every square of $R$ adjacent to $B$ shares exactly one edge with one of the squares of $B$. Let $s_1$ be a square adjacent to $B$. Let $t_1$ be the square in $B$ adjacent to $s_1$ and let $C_1$ be the corridor perpendicular to $B$ at $t_1$. Let $C_2$ be the corridor perpendicular to $C_1$ at $s_1$ and let $t_2$ be the square in $B$ and $C_2$.

Now, for $i \geq 2$, let $s_i$ be the square in $C_i$ adjacent to $t_i$, let $C_{i+1}$ be the corridor perpendicular to $C_i$ at $s_i$ and let $t_{i+1}$ be the square in $B$ and $C_{i+1}$.

We claim that for $i \geq 2$, $t_{i+1} \neq t_i$ and $t_{i+1}$ lies in $B$ between $t_{i-1}$ and $t_i$. Indeed, if $t_{i+1} = t_i$ then the corridor $C_{i+1}$ goes through $t_i$ and $s_i$, which are adjacent, and the corridor self-osculates, contradicting that there are no pathologies. As $C_{i+1}$ does not intersect $C_i$ twice and they already intersect at $s_i$, the square $t_{i+1}$ in $B$ must be between $t_i$ and $t_{i-1}$. Therefore, if $t_i$ and $t_{i+1}$ were not adjacent in $B$, then $t_{i+1}$ and $t_{i+2}$ get closer in $B$ than $t_i$ and $t_{i+1}$. For this reason, since $B$ is a corridor of finite length, there is some $N$ such that $t_N$ and $t_{N+1}$ are adjacent. Given that, by the assumption in this case, we have that $s_N \neq s_{N+1}$ and the corridor $C_{N+1}$ self-osculates (as it goes through $s_N$ and $s_{N+1}$ and these squares share a vertex). This gives a contradiction. $\square$

We use the previous result to show that pathologies can be eliminated from squared diagrams. The key ideas are to note that, in this setting, any type of (corridor) pathology contains a double intersection, that the internal area of such double intersection can be reduced to zero, and that this double intersection can be eliminated using hexagonal moves.

**Lemma 4.21.** *Let $D$ be a squared disc diagram. If $D$ contains a pathology $P$, then $\bar{P}$, the diagram enclosed by $P$, contains a double intersection.*

*Proof.* Suppose $P$ is not a double intersection. Take $s$ a square of $P$ with two opposite sides traversed by the fiber of the pathology. Consider the corridor perpendicular to the corridor of the pathology $P$ at $s$. Since it crosses the fiber of $P$ once, it must cross it a second time, which means it will intersect the corridor of the pathology at a second square. Thus, diagram $\bar{P}$ enclosed by $P$ contains a double intersection. $\square$

**Definition 4.22.** Let $D$ be a disc diagram over a prismatic presentation. Note that if $D$ contains two square 2-cells induced by the same relation (up to cyclic conjugation), and these two squares are glued along two consecutive edges, we can retesselate $D$ to a diagram $D'$ so that these two squares are replaced by two edges joined by a vertex. Similarly, if two square 2-cells induced by the same relation (up to cyclic permutation) are glued along three consecutive edges, we can replace the two squares by a single edge. This operation is what we call a *collapse.* Note that $a(D) = a(D') + 2$.

**Remark 4.23.** When we collapse two squares $s_1 \cup s_2$ in $D$ as described above, the pairs of edges along $\partial(s_1 \cup s_2)$ that get identified induced the same corridor before the collapse. Thus two edges in $\partial D$ induce a corridor if and only if they do after the collapse.

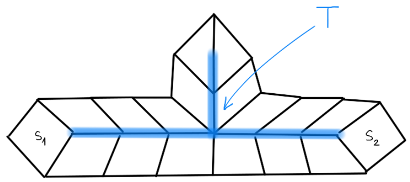
**Remark 4.24.** Let $D$ be a disc diagram over a prismatic presentation. As it happens with collapses, hexagonal moves preserve the corridors induced by the edges in $\partial D$. Simply observe that opposite edges in the boundary of a tessellated hexagon induce the same corridor before and after performing the hexagonal move.

**Lemma 4.25.** *Let $D$ be a disc diagram over a prismatic presentation. If $D$ contains a double intersection $P$ with internal area equal to zero, then $D$ admits a (possibly empty) sequence of hexagonal moves leading to a collapse which reduces the area of $D$.*

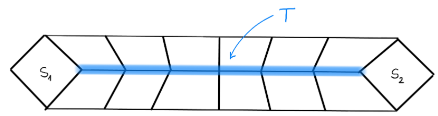
*Proof.* Let $C_1$ and $C_2$ be the two corridors which intersect twice (if they intersect more times, we restrict to two consecutive crossings). Let $s_1$ and $s_2$ be the squares at which $C_1$ and $C_2$ intersect each other. Because $P$ has internal area equal to zero, the fiber of $P$ contains no 2-cells in its interior region in the sense of Definition 4.11. Let us denote by $T$ the part of $\partial P$ that is contained in the interior region of the fiber of $P$ . Notice that $T$ is a subcomplex of the 1-skeleton of $D$, so it is a graph. Moreover, $T$ is connected because it can be seen as a retract of $P$, which is connected. Given that $D$ is simply connected, each loop in its 1-skeleton must be tesselated by 2-cells. Since $T$ lies inside the interior region of the fiber, it must be a connected graph with no loops, therefore a tree. The squares $s_1$ and $s_2$ are attached to $T$ at a single vertex, and the rest of squares of $P$ have exactly one side contained in $T$. Also, since all the edges of $T$ lie in the interior region of the diagram $D$, they are all adjacent to exactly two 2-cells (which must be 2-cells of $P$).

If $T$ is a single point, then $P$ consists of two squares glued along two consecutive edges. Due to the small pieces condition (**P2**) these squares must be induced by the same relation (up to cyclic conjugation), which means they admit a collapse. Suppose then that $T$ contains at least one edge that is a leaf edge. If the two squares of $P$ glued to this leaf are adjacent in $P$, then $D$ contains again two squares glued along consecutive edges, so the diagram admits a collapse (Fig. 14(A)). If this does not happen, it is necessarily because $T$ has only two leaves and $s_1$ and $s_2$ are glued to the vertices of these leaves. This means that $P$ has the form shown in Fig. 14(B) and it admits a series of hexagonal moves that yield a diagram with the same area and containing two squares glued along consecutive edges, which also implies that $D$ admits a collapse. ☐

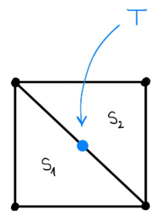
**Theorem 4.26.** *Let $D$ be a pathological squared disc diagram over a prismatic presentation. Then, $D$ admits a (possibly empty) sequence of hexagonal moves and a (non-empty) set of collapses that yields a diagram $D'$ with $\partial D = \partial D'$, with no*




(A) Two adjacent squares of $P$ glued to the same edge of $T$




(B) $s_1$ and $s_2$ at the two leaves of $T$




(C) $T$ is a single point

FIG. 14. Different configurations of $P$ in Lemma 4.25

*pathologies and satisfying that two edges in $\partial D'$ induce a corridor if and only if the corresponding edges in $\partial D$ do so.*

*Proof.* Let $D$ be a squared disc diagram over a prismatic presentation. If $D$ is pathology-free, we are done. Suppose $D$ contains pathologies. Then we can choose $P$ a pathology in $D$ satisfying that $\bar{P}$, the diagram enclosed by $P$, contains no pathologies different from $P$.

By Lemma 4.21, $P$ must be a double intersection of two corridors. Suppose that the internal area of $P$, say $a_i(P)$, is strictly greater that zero. Then, we may apply Theorem 4.20 taking $R$ to be the internal region of $P$ and $B$ to be the segment of corridor that remains when we remove from $P$ one of the corridors involved in the double intersection (see Figure 15). Indeed, $R \sqcup B$ has $B$ as an arc at its boundary and $a(R) = a_i(P) > 0$. Moreover, $R \sqcup B$ contains no pathologies and $B$ is outermost boundary arc in $R \sqcup B$ because $\bar{P}$, the diagram enclosed by $P$, contains no pathologies different from $P$. Therefore, by this theorem, we conclude that the internal area of $P$ can be reduced without modifying the area of $D$. This means that we may now assume that the internal area of $P$ is zero. So, by Lemma 4.25, the diagram admits a sequence of hexagonal moves leading to a collapse that reduces yields a diagram $D'$ with smaller area. If the new diagram $D'$ has no pathologies, we are done; if not, we repeat the reasoning recursively. Since in each iteration the area of the diagram decreases 2 units and the initial diagram has finite area, this process will eventually finish with a diagram of area zero or area one (which are always non pathological) or with a diagram of greater area which is non pathological. Note that, because the retessellations only involve hexagonal moves and collapses, Remarks 4.23 and 4.24 ensure that two edges in the boundary of the final diagram induce a corridor if and only did they did in the original one. This yields the desired, non-pathological diagram.

$\square$

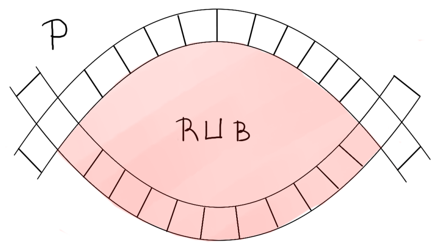



FIG. 15. Choosing the subcomplexes $R$ and $B$ in the double intersection from Theorem 4.20.

**Theorem 4.27.** *Let $D$ be a disc diagram over a prismatic presentation. Then, $D$ cannot contain spiders with circular legs or with intersecting legs.*

*Proof.* If $D$ has siamese spiders, these can be removed by Lemma 4.13. Therefore we can assume that $D$ has no siamese spiders. Assume that $D$ exhibits pathologies as in the statement, choose among them a pathology $P$ so that $\bar{P}$ does not contain any other circular legs or intersecting legs. If there are any heads in the interior of $P$, apply prismatic moves to remove them. Let us now consider the two possible cases.

Assume that $P$ is given by a pair of intersecting legs $L_1$, $L_2$ and a head $H$. We can retessellate the squared subdiagram given by $\overline{P} - H$ using Theorem 4.26. Note that the retessellated subdiagram is now free of pathologies and the boundary edges of $L_1$ and $L_2$ still define intersecting legs $L_1'$ and $L_2'$. We may now apply Theorem 4.20 on the retessellated subdiagram taking $R$ to be the interior region of $P$ and $B$ to be $L_2'$: indeed, $L_2'$ is an outermost boundary arc, if it was not there would be a double intersection between $L_1'$ and another corridor. The application of Theorem 4.20 leaves us with a diagram where the pathology given by the two intersecting legs $L_1'$ and $L_2'$ has an interior region with area zero. $\overline{P}-(L_1 \cup H)$ using Theorem 4.26. Note that the retessellated subdiagram is now free of pathologies and the boundary edges of $L_2$ still define a leg $L_2'$ which intersects the leg $L_1$. We may now apply Theorem 4.20 on the retessellated subdiagram taking $R$ to be the interior region of $P$ and $B$ to be $L_2'$, which leaves us with a diagram where the pathology given by these two intersecting legs has an interior region with area zero.

Arguing as in Lemma 4.25 we take the corresponding tree $T$. If any leaf edges in $T$ correspond to pairs of squares glued along two edges, due to **P2** the two squares must be induced by the same relation and can be collapsed, trimming this way the corresponding leaf edge in the tree $T$. After a finite number of iterations, the tree must be a line and the leaf edge which is not adjacent to the head $H$ must correspond to the intersection square of the two legs. At this point, a sequence of hexagonal moves can be performed to reach a diagram where the intersection square is glued to the head along two consecutive edges, contradicting **P2**. See Fig. 16.

We now know that $P$ could not be given by a pair of intersecting legs. Assume that $P$ is given by a head $H$ and a circular leg $L$ with no heads in its interior. Let $s$ be one of the squares of $L$ that is adjacent to the head. We can retessellate the squared subdiagram given by $\bar{P} - (s \cup H)$ using Theorem 4.26 in order to eliminate pathologies from this subdiagram while maintaining $L$. In the retessellated

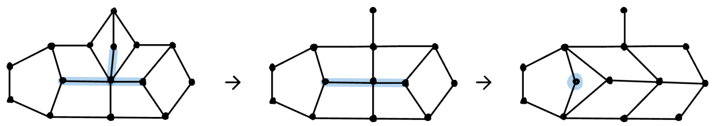


FIG. 16. Trimming tree process for two intersecting legs.

subdiagram, we may apply Theorem 4.20 taking $R$ to be the interior region of the pathology and $B$ to be the retessellated leg $L$. This gives a diagram where the interior region of the pathology has area zero, we take the corresponding tree $T$ as in Lemma 4.25. Because this pathology only involves one corridor, all of the leaf edges in this tree that are not adjacent to the head $H$ of the pathology must correspond to two squares glued along two consecutive edges. Due to **P2**, these pairs must come from the same relation and a collapse can be performed. After a finite number of collapses we end up with a tree consisting of one edge one of whose vertices belongs to the head $H$. Now the circular leg is exhibited by the subdiagram given by the head and the adjacent two squares that are glued along two consecutive edges (Fig. 17). Applying one last collapse to the pathology shows that the head is induced by a relation which was not cyclically reduced, contradicting Definition 3.6. We conclude that a disc diagram over a prismatic presentation cannot exhibit circular legs either.

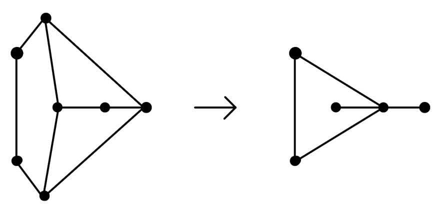


FIG. 17. Collapse of degenerate circular leg gives a not cyclically reduced head.

□

**Corollary 4.28.** *Let $D$ be a disc diagram over a prismatic presentation. There is an alternative diagram $D'$ with $\partial D = \partial D'$ such that $D'$ contains no pathologies, spider pathologies or siamese spiders.*

*Proof.* Using Theorem 4.27, we know that $D$ contains no spider pathologies. Applying Lemma 4.13, siamese spiders may also be removed from $D$. If the resulting diagram has no pathologies, we are done. Otherwise, take a pathology $P$ and remove any heads in its interior region using prismatic moves. This shows that the set of retessellations of $D$ with no siamese spiders, spider pathologies and containing a pathology $P$ with no heads in its interior region is non-empty. Among all of these diagrams, choose one with minimal area and call it $D'$. Let $P'$ be a pathology in $D'$ with no heads in its interior region. Using again Theorem 4.27 we know that

$P'$ is a corridor pathology and $\overline{P'}$ is a squared diagram, which means that Theorem 4.26 can be applied to it reducing the area of the pathology and contradicting the minimality in the choice of $P'$. □

## 5. Squashing spiders and FFTP

**Definition 5.1.** We say that $D$ has a spider at its boundary if it contains a spider's head which intersects the boundary of $D$.

In order to show that groups with a prismatic presentation have FFTP, this section focuses on characterizing non-geodesic words over the generators of such a presentation. When looking at a disc diagram with a non-geodesic word $w$ along its boundary, we can informally say that this non-geodesicity translates into the presence of a corridor or a spider that has "too many" legs ending in $w$ (this is formalized in the notion of $w$-arc, see Definition 5.3). To make sure that there is a shorter path fellow-travelling with $w$, one wants to find an equivalent disc diagram in which this corridor or spider is "squashed" against the boundary. This idea will be formalized throughout this section, leading to a characterization of geodesics (Theorem 5.21) and a proof that these groups satisfy FFTP (Theorem 5.24).

**Definition 5.2** (Spider arc)**.** Let $D$ be a disc diagram over a prismatic presentation. Given a non-pathological spider in $D$, a *spider arc* is given by the union of its head $H$ together with two of its legs. The two edges where these legs intersect $\partial D$ are called the *extremes* of the spider arc. Given a spider arc with extremes $e$, $e'$, consider its 1-skeleton and remove from it the edges in the equivalence classes of $e$ and $e'$. Since $D$ embeds on the plane we obtain two 1-dimensional complexes that we call the *sides* of the spider arc.

**Definition 5.3** ($w$-arc)**.** Let $D$ be a disc diagram over a prismatic presentation containing no spider pathologies or siamese spiders and let $w$ be a subword of the word labelling $\partial D$. We say that a subdiagram of $D$ is a *$w$-arc* if it is either an arc or a spider arc whose extremes lie in $w$ and, in the case of a spider given by an $n$-gonal head, $\lfloor \frac{n-2}{2} \rfloor$ of its legs end in $w$ between the extremes of the spider arc. In particular, if the $w$-arc is given by a spider the majority of its legs end in $w$.

The *sides* of a $w$-arc are the sides of the corresponding arc or spider arc.

**Definition 5.4** (Region enclosed by $w$-arc)**.** Given a $w$-arc $A$, the *region enclosed* by $A$ and $\partial_w D$, denoted $I_A$, is the region enclosed by $\partial_w D$ and the side of $A$ whose end vertices are closest in $\partial_w D$ along $w$. Informally, we choose the region $I_A$ so that it does not include $A$.

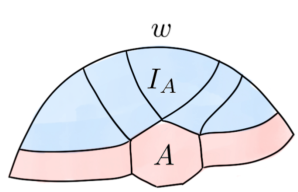
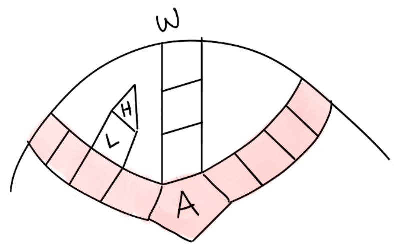


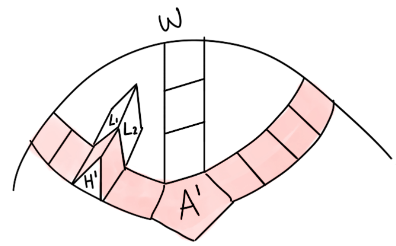

Fig. 18. Spider $w$-arc $A$ in a disc diagram $D$, the majority of the spider's legs end in $w$. The region enclosed by $A$ and $\partial_w D$ is denoted $I_A$.

**Remark 5.5.** A $w$-arc $A$ in a disc diagram $D$ is uniquely determined by the two edges in $w$ where $A$ intersects $\partial_w D$.

**Definition 5.6** (Outermost $w$-arc)**.** Given a diagram $D$ as in Definition 5.3 and a word $w$ along $\partial D$, we say that a $w$-arc $A$ is *outermost* in $D$ if $I_A$ does not contain other $w$-arcs.




(A) A $w$-arc containing a spider head $H$.




(B) $H$ can be pushed out to $H'$ using a prismatic move.

FIG. 19. Pushing heads out of a $w$-arc, retesselating the half-prism $H \cup L$ to its other half $H' \cup L_1 \cup L_2$.

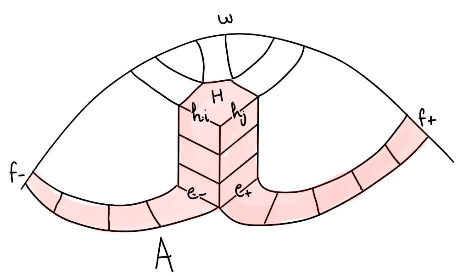
The following two statements show that the set of $w$-arcs in a diagram is preserved by both hexagonal and prismatic moves, and their relative containment is preserved in the sense that outermost $w$-arcs map to outermost $w$-arcs.

**Lemma 5.7** (Prismatic moves preserve $w$-arcs)**.** *Consider a disc diagram $D$ over a prismatic presentation with no spider pathologies or siamese spiders and a word $w$ along $\partial D$, and let $D'$ be a disc diagram obtained from $D$ by applying a prismatic move. Two edges $f_+, f_-$ in $w$ determine a $w$-arc in $D$ if and only if they determine a $w$-arc in $D'$.*

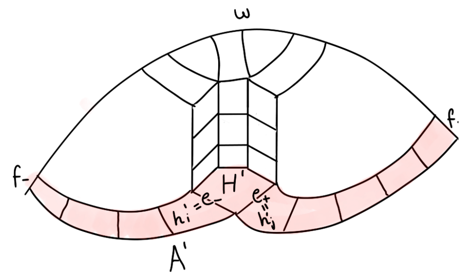
*Proof.* Let $H \cup L_1 \cup \ldots \cup L_n$ be the half-prism in $D$ to which the prismatic move is applied, where $H$ denotes a polygon and $L_1, \ldots L_n$ denote the $n$ lateral faces. The prismatic move will retesselate the half-prism to its other half $H' \cup L_{n+1} \cup \ldots \cup L_m$ where $H'$ is its other base and $L_{n+1} \cup \ldots \cup L_m$ are the remaining lateral faces. For $i = 1, \ldots, m$, denote by $h_i$, $h_i'$ the edges of $H$ and $H'$ respectively that are connected by the lateral face $L_i$ in the full prism.

Consider a $w$-arc $A$ in $D$ with extremes $f_-$ and $f_+$ in $\partial_w D$. Suppose that $A$ traverses the original half-prism and call $e_-, e_+$ the two intersection edges in the boundary of the half-prism. If $A$ intersects $H$ in edges $h_i, h_j \in H$ as in Fig. 20(A), the prismatic move will yield a new disc diagram $D'$ where edges $f_-, f_+$ still determine a $w$-arc $A'$ which traverses $e_-$ and $e_+$ and goes through the other base $H'$ through its edges $h_i'$ and $h_j'$ (Fig. 20(B)). This is a consequence of the fact that the prismatic move can be understood as an elongation or contraction of the legs (lateral faces) of a spider with head $H$.

Keeping the above notation, consider now the case where the $w$-arc $A$ traverses the half-prism but not $H$. This means that $A$ is transversal to the lateral faces $L_1, \ldots, L_n$ as in Fig. 21(A), and in this case the prismatic move yields a diagram $D'$ where $f_-, f_+$ still determine a $w$-arc $A'$ that traverses $e_-, e_+$ and is transversal to the remaining lateral faces $L_{n+1}, \ldots, L_m$ as Fig. 21(B) shows.

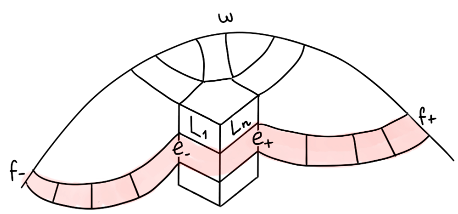



(A) In $D$, the $w$-arc $A$ induced by $f_-, f_+$, crossing the half-prism in $e_-$ and $e_+$ and intersecting the head $H$ through $h_i$ and $h_j$.

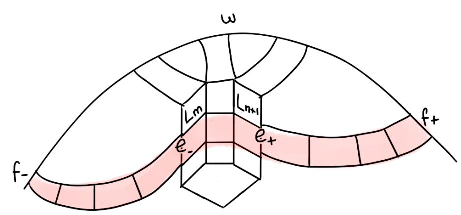



(B) In $D'$, the $w$-arc $A'$ induced by $f_-, f_+$, crossing the half-prism in $e_-$ and $e_+$, and intersecting the head $H'$ through $h_i', h_j'$.

Fig. 20. On the left, a diagram $D$ with a $w$-arc $A$ traversing the base of a half-prism. A prismatic move yields the diagram $D'$ on the right where $A$ becomes $A'$ and traverses the other base.




(A) A $w$-arc $A$ induced by $f_-, f_+$, crossing the half-prism in $e_-$ and $e_+$ through legs $L_1, \ldots, L_n$.




(B) A $w$-arc $A'$ induced by $f_-, f_+$, crossing the other half-prism in $e_-$ and $e_+$ and traversing $L_{n+1}, \ldots, L_m$.

Fig. 21. On the left, a diagram $D$ with a $w$-arc $A$ traversing the lateral faces of a half-prism. A prismatic move yields the diagram $D'$ on the right where $A$ becomes $A'$ and traverses the other lateral faces.

The reasoning above shows that edges $f_-, f_+$ will only determine a $w$-arc before [after] the prismatic move if they do so afterwards [before]. Moreover, any $w$-arcs in $D$ not traversing the retesselated half-prism will not be affected by the prismatic move. $\square$

**Lemma 5.8.** *Consider a disc diagram $D$ over a prismatic presentation with no spider pathologies or siamese spiders and a word $w$ along $\partial D$, and let $D'$ be a disc diagram obtained from $D$ by applying a prismatic move. If $B$ and $C$ are $w$-arcs in $D$ inducing $w$-arcs $B'$, $C'$ in $D'$ that satisfy $C \subset I_B$, then it cannot happen that $B' \subset I_{C'}$.*

*Proof.* Let us see why the containment between $w$-arcs cannot be inverted by prismatic moves. Suppose $B$ and $C$ are $w$-arcs in $D$ inducing $w$-arcs $B'$, $C'$ in $D'$ that satisfy $C \subset I_B$. Let $f_-, f_+$ and $g_-, g_+$ be the edges in $\partial D = \partial D'$ that determine the arcs $B, B'$ and $C, C'$ respectively. Because $C \subset I_B$, in $\partial_w D$, $g_-$ and $g_+$ lie between the edges $f_-$ and $f_+$. Since prismatic moves do not change the boundary of $D$ and $B'$ and $C'$ are determined by the same edges of $\partial_w D$ as $B$ and $C$, it

**LAM:** When editing this I realised that the lemmas regarding hexagonal moves are not used anywhere, I think they should have been used somewhere? Hemos comentado por ahora los lemas de "hexagonal moves preserve $w$-arcs and do not change containment".

cannot happen that $B' \subset I_{C'}$ (for this to happen, we would need to have that $f_-$ and $f_+$ lie between the edges $g_-$ and $g_+$ in $\partial_w D$). $\square$

The next lemma shows that we can retesselate the region enclosed by an outermost $w$-arc and the boundary $\partial_w D$ so that this region does not contain any spider heads or spider pathologies and the initial $w$-arc remains outermost.

**LAM:** Nota: aquí hemos quitado la hipótesis de corridor pathologies porque creemos que no hace falta... lo dejo escrito por si acaso

**Lemma 5.9.** *Let $D$ be a disc diagram over a prismatic presentation containing no spider pathologies and no siamese spiders. Let $w$ be a word along the boundary of $D$. If $D$ contains an outermost $w$-arc $A$, then there exists an alternative disc diagram $D'$ satisfying the following:*

*(1) $\partial D = \partial D'$,*
*(2) $D'$ contains an outermost $w$-arc $A'$ with the same extremes as $A$,*
*(3) the region $I_{A'}$ enclosed by $A'$ and $\partial_w D$ contains no spider heads (that is, it is a squared diagram),*
*(4) if the $w$-arc $A'$ is given by a spider, its head intersects $\partial_w D$, and*
*(5) the region $I_{A'} \cup A'$ contains no corridor or spider pathologies.*

*Proof.* Let $D = D_0$ be as in the statement of the theorem. If $I_A$ contains heads, we may apply prismatic moves in order to take them out: indeed, since $D$ contains no spider pathologies and $A$ is outermost, every head $H$ in $I_A$ has a leg that intersects $A$, call it $L$. This head can be removed from the region enclosed by the $w$-arc by applying a prismatic move to $H \cup L$ in $A \cup I_A$ and, doing this iteratively for every head inside $I_A$, we obtain a new diagram $D_1$ satisfying (1), (2) and (3). Notice that (2) is satisfied by Lemmas 5.7 and 5.8.

Let $A_1$ be the outermost $w$-arc in $D_1$ having the same extremes as $A$. If $A_1$ is given by a spider with head $H_1$, we can always perform a prismatic move along one of the legs of the spider arc so as to push the head towards $\partial_w D$. In this way, we obtain a diagram $D_2$ with outermost $w$-arc $A_2$ satisfying (1), (2), (3) and (4). As before, (2) remains true by Lemmas 5.7 and 5.8.

Notice that $D_2$ contains no spider pathologies by Theorem 4.27. It may happen, however, that the prismatic moves that we have performed have created corridor pathologies in $(I_{A_2} \cup A_2) - H_2$, where $H_2$ is the head of $A_2$ (if $A_2$ is a spider arc). If this is the case, we retessellate $(I_{A_2} \cup A_2) - H_2$ using Theorem 4.26 to obtain a new diagram $D_3$. Observe that, because this retessellation does not change the corridors induced by pairs of edges in the boundary, $D_3$ contains an outermost $w$-arc $A_3$ induced by the same boundary edges as $A_2$. Now, because $I_{A_3} \cup A_3$ is the union of $(I_{A_2} \cup A_2) - H_2$ (which does not contain corridor pathologies) and the head $H_2$, and taking into account Theorem 4.27, we see that $D' = D_3$ satisfies all five properties in the statement of the theorem. $\square$

Our next goal is to see that, given a disc diagram with an outermost $w$-arc as in Lemma 5.9 above, we can transform it so as to "erase" the area enclosed above the $w$-arc. We will first show how this is possible if the $w$-arc is given by a spider and then if it is a corridor.

**Definition 5.10.** Let $w$ be a word over the standard generators of prismatic presentation and consider a disc diagram $D$ with an outermost $w$-arc $A$ as in Lemma 5.9. If $A$ is a corridor $w$-arc, we say that $A$ is *empty* if the area of $I_A$ is null. If $A$ is a spider $w$-arc given by a spider $S$, we say that $A$ is *empty* if the area of $I_A - S$ is null.

We will see that a diagram containing a $w$-arc admits retessellations that empty this $w$-arc (Theorem 5.15). To do so, we first focus on the case where the $w$-arc is given by a spider. We establish the following notation:

**Notation 5.11.** Let $D$ be a disc diagram over a prismatic presentation and let $w$ be a word along its boundary. Suppose that $D$ has an outermost, spider $w$-arc $A$ as described in Lemma 5.9. We introduce the following notation to describe the spider (see Fig. 22).

We denote by $w_0$ the (possibly trivial) subword of $w$ that labels $H \cap \partial_w D$ (it is not difficult to see, from the definition of arc, that $H \cap \partial_w D$ is a non-empty, connected subcomplex of $\partial_w D$). If $w_0$ is given by a single letter we may also refer to it as $L_0$ referencing the (trivial) leg this edge represents. We may now write $w = w' w_0 w''$ and we can further factorise

$$\begin{aligned} w' &= w_{-(p+1)} f_{-p} w_{-p} f_{-(p-1)} \cdots f_{-2} w_{-2} f_{-1} w_{-1}, \text{ and} \\ w'' &= w_1 f_1 w_2 f_2 \cdots w_q f_q w_{q+1}, \end{aligned}$$

where each $f_i$ is an edge where one of the spider's legs intersects $\partial_w D$ and $w_i$ are the remaining subwords in the partition induced by the edges $f_i$. We denote by $L_i$ the leg of the spider which has $f_i$ as an extremal edge and, for $i \in \{1, \dots, q\}$, the side of $L_i$ that shares a vertex with $w_i$ will be called the upper side $u_i$ and the side of $L_i$ that shares a vertex with $w_{i+1}$ will be called the lower side $l_i$. Symmetrically for $i \in \{1, \dots, p\}$, the side of $L_{-i}$ that shares a vertex with $w_{-i}$ will be called $u_{-i}$ and the side of $L_{-i}$ that shares a vertex with $w_{-(i+1)}$ will be called $l_{-i}$.

The region enclosed by $u_1$ and $w_1$ is denoted $R_1$ and, for $i \in \{2, \dots, q\}$, the region enclosed by $u_i$, $w_i$ and $l_{i-1}$ is denoted $R_i$. Symmetrically, the region enclosed by $u_{-1}$ and $w_{-1}$ is denoted $R_{-1}$ and, for $i \in \{2, \dots, p\}$, the region enclosed by $u_{-i}$, $w_{-i}$ and $l_{-(i-1)}$ is denoted $R_{-i}$. We convene that $R_0$ is the region enclosed by $w_0$ and $\partial_w D$, which has null area. The notation is clarified in Fig. 22.

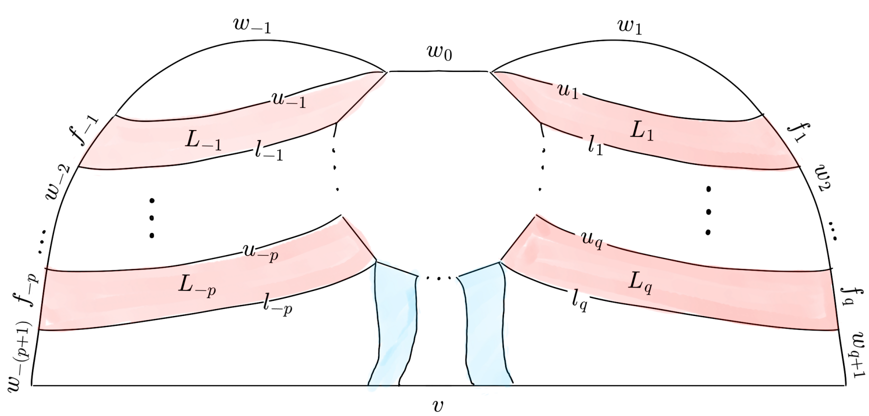



FIG. 22. Spider $w$-arc in a disc diagram $D$ for a word $wv^{-1}$ where the spider's head intersects $\partial_w D$: each leg $L_i$ in the $w$-arc as an upper side $u_i$ and a lower side $l_i$, the end edges of the legs break the path $w$ into a concatenation of $f_i$-edges and $w_i$-paths, and the region enclosed by $w_i, u_i$ and $l_{i-1}$ is denoted by $R_i$.

**Definition 5.12.** Given $n, m \in \mathbb{Z}$ with $m < n$, a word $v$ over the standard generators of a prismatic presentation representing the trivial element and a subword $w$ of $v$, we define the set $\mathcal{D}_{m,n}(w, v)$ as the set of disc diagrams for $v$ with an outermost spider $w$-arc $A$ as in Lemma 5.9 whose regions $R_i$ have null area for $m \leq i \leq n$ (Notation 5.11). For $n \geq 0$ we use the notation $\mathcal{D}_n$ as a shorthand for $\mathcal{D}_{0,n}$ and for $m < 0$ we use $\mathcal{D}_m$ as a shorthand for $\mathcal{D}_{m,0}$.

**Proposition 5.13.** *Let $v$ be a word over the standard generators of a prismatic presentation representing the trivial element and let $w$ be a subword $v$. Suppose that $\mathcal{D}_0(w, v)$ is non-empty, that is, there is a diagram $D$ with an outermost spider $w$-arc $A$ as in Lemma 5.9, and that this spider has legs $L_{-p}, \dots, L_{-1}, L_1, \dots, L_q$ intersecting $\partial_w D$ as in Fig. 22. The $w$-arc $A$ (and therefore the diagram $D$) can be sequentially altered to obtain a diagram $D''$ with a $w$-arc $A''$ satisfying the requirements for $D''$ to belong to $\mathcal{D}_{-p,q}(w, v)$. In particular, if $\mathcal{D}_0(w, v)$ is non-empty then $\mathcal{D}_{-p,q}(w, v)$ is non-empty.*

*Proof.* We argue inductively. If the diagram $D$ does not belong to $\mathcal{D}_q(w, v)$, let $n$ be the smallest integer such that the region $R_n$ of the $w$-arc $A$ has non-zero area.

We will see that $D$ admits a hexagonal move that reduces the area of $R_n$. The subcomplex $D_n \subset D$ given by $R_n \cup L_n$ is a squared diagram that has the leg $L_n$ as an outermost boundary arc, contains no pathologies and none of its corridors begins and ends in $\partial D_n \cap R_n$ – this is due to $A$ being outermost and $A \cup I_A$ exhibiting no pathologies. Theorem 4.20 allows us to push a square of $R_n$ through $L_n$, retessellating $D_n$ to $D'_n \cup s$ where:

1. The subcomplex $D'_n$ satisfies $D'_n = L'_n \cup R'_n$ with $L'_n$ being an arc at its boundary,
2. The subcomplex $L'_n$ is a spider leg induced by the same extremal edges as the leg $L_n$,
3. The area $a(R'_n)$ is less than $a(R_n)$, and
4. The retessellation $D_n$ to $D'_n \cup s$ induces a retessellation $D$ to $D'$ where the $w$-arc $A$ gets transformed to a $w$-arc $A'$ and the square $s$ belongs now to the region $R'_{n+1}$ of $A'$.

Note that, since this retessellation only affects the leg $L_n$ and this leg is preserved, the new arc $A'$ is given by a spider with the same leg structure as $A$, say $L'_{-p}, \dots, L'_{-1}, L'_1, \dots, L'_q$. Furthermore the retessellation introduces no pathologies or $w$-arcs, so $A'$ satisfies the conditions for $D'$ to belong to $\mathcal{D}_n(w, v)$.

By repeated application of Theorem 4.20 we can build a disc diagram starting from $D$ that contains an outermost $w$-arc which, having the same leg structure as $A$, ensures that the disc diagram belongs to $\mathcal{D}_q(w, v)$. We can now proceed symmetrically on the remaining ("negative") legs to transform this diagram into a new one $D''$ with an outermost $w$-arc $A''$ that, preserving the leg structure of $A$, satisfies the conditions for $D''$ to belong to $\mathcal{D}_{-p,q}(w, v)$. $\square$

The above result shows how to progressively empty a spider $w$-arc, we do the same with a corridor $w$-arc.

**Proposition 5.14.** *Let $D$ be a disc diagram over a prismatic presentation and let $w$ be a subword of $\partial D$. Suppose that $D$ has an outermost $w$-arc $A$ which is a corridor and whose interior region $I_A$ is squared and free of pathologies. If $D$ minimises the area of $I_A$, then $A$ must be an empty $w$-arc.*

*Proof.* If $a(I_A) > 0$, we may apply Theorem 4.20 taking $R = I_A$ and $B = A$. Indeed, given that the $w$-arc $A$ is outermost, there are no corridors beginning and ending at $\partial(I_A \sqcup A) \cap I_A$, so all the hypotheses of the theorem are satisfied. Thus, $I_A \sqcup A$ admits a hexagonal move that yields a retessellation $D' \sqcup s$ of $I_A \sqcup A$ where there is a new outermost $w$-arc $A'$ satisfying that $I_{A'}$ is squared and free of pathologies and satisfies that $a(I_{A'}) < a(I_A)$, contradicting the minimality assumption. □

Now, we conclude:

**Theorem 5.15.** *Let $D$ be a disc diagram over a prismatic presentation and let $w$ be a subword of $\partial D$. If $D$ contains an outermost $w$-arc, there is an alternative disc diagram $D'$ with $\partial D = \partial D'$ containing an empty $w$-arc. Moreover, if $D$ is free of siamese spiders, the empty $w$-arc in $D'$ is induced by the same end-edges as the original one.*

**LAM:** Sutileza: creo que esto es cierto si pedimos free of siamese spiders *solo en la interior region*, pero esto no es lo que el enunciado de 5.9 pide – en principio dejo el statement menos preciso pero más fácil de deducir de 5.9

*Proof.* If in $D$ contains siamese spiders, these can be removed using Lemma 4.13. The original $w$-arc might not be outermost after the retessellation, but in this case a new outermost $w$-arc could be found in its interior region. Now, the statement is a direct consequence of Lemma 5.9, Propositions 5.13 and 5.14. □

**Definition 5.16** (Shortcut for empty corridor $w$-arc)**.** Let $D$ be a disc diagram over a prismatic presentation and let $w$ be a subword of $\partial D$. Suppose $D$ has an empty corridor $w$-arc $A$. Let $e, e'$ be the edges in the equivalence class of the corridor that are contained in $\partial_w D$, let $u_A$ be the side of the corridor running along $\partial_w D$ and let $l_A$ be the other side of the corridor $A$. For the subword of $w$ given by

$$w_A = e \cdot u_A \cdot e'$$

we define the *shortcut for $w_A$ through $A$*, denoted $\alpha_A$, to be the path with the same endpoints as $w_A$ given by $l_A$.

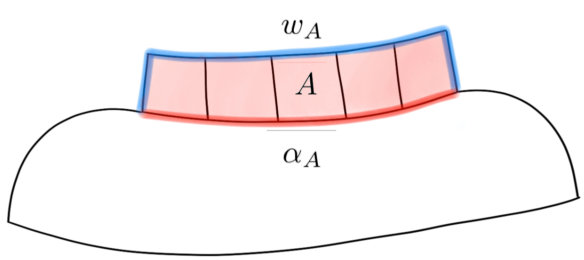



FIG. 23. Shortcut for an empty corridor $w$-arc $A$.

**Proposition 5.17.** *Let $D$ be a disc diagram over a prismatic presentation and let $w$ be a subword of $\partial D$. Given an empty corridor $w$-arc $A$ in $D$, the shortcut $\alpha_A$ for $w_A$ is strictly shorter than $w_A$ and 1-fellow travels with it.*

**Definition 5.18** (Shortcut for empty spider $w$-arc)**.** Let $D$ be a disc diagram over a prismatic presentation and let $w$ be a subword of $\partial D$. Let $A$ be an empty spider $w$-arc coming from a spider with head $H$ and whose legs $L_{-p}, \ldots, L_q$ intersect $\delta_w D$ as in Notation 5.11. For the subword $w_A$ of $w$ given by

$$w_A = f_{-p} w_{-p} \ldots w_0 \ldots w_q f_q,$$

we define the *shortcut for $w_A$ through $A$*, denoted $\alpha_A$, to be the path with same endpoints as $w_A$ given by the concatenation $l_{-p} \cdot h_{-p,q} \cdot l_q$ where $l_{-p}$ and $l_q$ are the lower sides of the legs $L_{-p}$ and $L_q$ respectively and $h_{-p,q}$ is the shortest path along the 1-skeleton of $H$ between $l_{-p}$ and $l_q$.

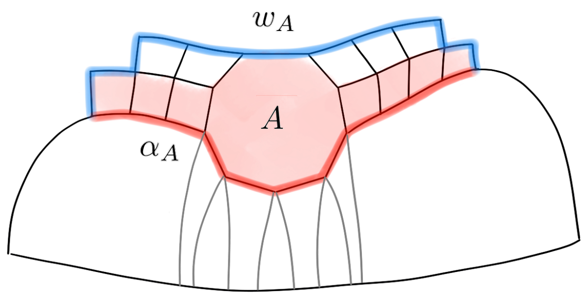



FIG. 24. Shortcut for an empty spider $w$-arc $A$.

**Proposition 5.19.** *Let $D$ be a disc diagram over a prismatic presentation and let $w$ be a subword of $\partial D$. Suppose $D$ has an empty spider $w$-arc $A$ whose head $H$ has $n$ edges, and following the definition of the subwords $w_i$ given in Notation 5.11, there are bijections between $\cup_{i<0}\{w_i\}$ and $l_{-p}$ and between $\cup_{i>0}\{w_i\}$ and $l_q$. In particular, the shortcut $\alpha_A$ is shorter than the path $w_A$.*

*Proof.* We define a map

$$\pi\colon \bigcup_{i\in\{-p,\ldots,q\}\setminus\{0\}} \{w_i\} \to l_{-p} \cup l_q$$

that assigns, to each edge $e$ along the domain, the corresponding edge $p(e)$ along $l_{-p} \cup l_q$ in its equivalence class.

We first argue that such edge exists. If the equivalence class of $e$ does not intersect an edge in $l_{-p} \cup l_q$ it must intersect either $w_A$ or the head $H$ of the spider $w$-arc, but none of these options are possible because of $A$ being outermost and the edge $e$ not being one of the $f_i$-edges.

Moreover, $\pi$ is bijective. If it was not injective and for two edges $e$, $e'$ in the domain we had $\pi(e) = \pi(e')$, it would mean that $e$ and $e'$ span the same corridor contradicting the fact that $A$ is outermost. To see that it is surjective, take an edge $e$ along $l_{-p} \cup l_q$. If the corridor spanned by $e$ does not contain an edge in the domain of the map $\pi$ it must contain another edge in $l_{-p} \cup l_q$ or in the head $H$ of the spider, but this is not possible because it would yield either an intersection between two legs of a spider or a double intersection between a leg and the corridor spanned by $e$. Notice that, because $A \cup I_A - H$ is disconnected, no corridor induced by an edge of $l_{-p}$ (respectively, $l_q$) can contain an edge in $w_i$ with $i > 0$ (respectively, $i < 0$). As a consequence, the restrictions of $\pi$ to $\cup_{i<0}\{w_i\}$ and $\cup_{i>0}\{w_i\}$ give us the bijections of the statement.

To conclude that the shortcut $\alpha_A$ is shorter than $w_A$ using the previous bijection, just notice that the remaining edges in $w_A$ (given by the $f_i$-edges and the $w_0$-edges) add up to $\lfloor \frac{n}{2} \rfloor + 1$ edges and that the remaining edges in $\alpha_A$ (given by the path $h_{-p,q}$) add up to at most $\lfloor \frac{n}{2} \rfloor$ edges. $\square$

**Proposition 5.20.** *Let $D$ be a disc diagram over a prismatic presentation and let $w$ be a subword of $\partial D$. If $D$ is an empty spider $w$-arc $A$ whose head has $n$ edges, the shortcut $\alpha_A$ $k$-fellow travels with $w_A$, where $k \leq \lfloor \frac{n}{2} \rfloor + 1$.*

*Proof.* Given that the spider is empty, all vertices belonging to $w'$ are at distance at most $p$ from a vertex belonging to $l_{-p}$, all vertices in $w_0$ are at most at distance $\lfloor \frac{n}{2} \rfloor$ from a vertex in $h_{-p,q}$ and all vertices in $w''$ are at distance at most $q$ from a vertex in $l_q$. The same bounds are true if we consider vertices in $\alpha_A$ and search for neighbours in $w_A$. Moreover, the three bounds are lower or equal than $k = \lfloor \frac{n}{2} \rfloor + 1$, which can be taken as a constant for the FFTP property. □

The following result characterises geodesics over the generators of a prismatic presentation.

**Theorem 5.21** (Geodesic characterisation)**.** *Let $w$ and $u$ be words over the generators of a prismatic presentation representing the same group element. Then, $w$ is geodesic if and only if no disc diagram for $wu^{-1}$ that is free of pathologies and siamese spiders contains a $w$-arc.*

*Proof.* Assume $w$ is geodesic and, for a contradiction, assume that there is a word $u$ with $u =_G w$ for which $wu^{-1}$ admits a disc diagram free of pathologies and siamese spiders that contains a $w$-arc. By Theorem 5.15, this $w$-arc can be assumed to be empty and Propositions 5.17 and 5.19 give then a path shorter than $w$ with the same endpoints.

Suppose now that $w = x_1x_2 \cdots x_n$ is not geodesic, then there is a word $u = y_1y_2 \cdots y_m$ such that $w =_G u$, $n > m$ and $x_i, y_j$ are generators of the trickle group or their inverses. Let $D$ be a disc diagram for $wu^{-1}$ with no pathologies or siamese spiders. Let $C_1, C_2, \dots, C_k$ be the corridors in $D$ which intersect $\partial D$ twice, together with the spiders in $D$ (all of whose legs end in $\partial D$ because there are no siamese spiders). Each edge in the boundary of $D$ is an extreme of some $C_i$. For every $i \in \{1, \dots, k\}$, if we denote by $n_i$ the number of edges in $\partial_w D$ which are extremes of $C_i$ and by $m_i$ the number of edges in $\partial_u D$ which are extremes of $C_i$, the following equalities must hold:

$$\begin{aligned} n &= n_1 + n_2 + \cdots + n_k, \\ m &= m_1 + m_2 + \cdots + m_k. \end{aligned}$$

If $n_i \leq m_i$ for all $i \in \{1, \dots, k\}$, then $n = n_1+n_2+\cdots+n_k \leq m_1+m_2+\cdots+m_k = m$, which is a contradiction. So there exists $j \in \{1, \dots, k\}$ such that $n_j > m_j$. If $C_j$ is a spider, the conclusion is obvious. If $C_j$ is a corridor, it also holds that $n_j + m_j = 2$, so it must be that $n_j = 2$ and $m_j = 0$, and $C_j$ crosses $\partial_w D$ twice. Therefore, $D$ contains a $w$-arc. □

**Remark 5.22.** The above proof shows that something slightly stronger than Theorem 5.21 is true. Indeed, taking $w$ and $u$ as above, the first paragraph of the proof remains true if the disc diagram for $wu^{-1}$ containing a $w$-arc is chosen arbitrarily – we can drop the hypothesis that the disc diagram is free of pathologies and siamese spiders.

**Corollary 5.23.** *Let $\langle \Sigma | H \sqcup S \rangle$ be a prismatic presentation and let $w$ be a non-geodesic word over the generating set $\Sigma$. There exists a shorter word $u$ representing the same group element as $w$ such that $u$ asynchronously $k$-fellow travels with $w$, where $k = 1$ if $H$ is empty and $k = \lfloor \frac{n}{2} \rfloor + 1$ otherwise, for $n$ a bound on the length of the relators in $H$.*

*Proof.* If $w$ is non-geodesic, take $v$ to be a shorter word over $\Sigma$ representing the same group element as $w$. By Theorem 5.21 there is a diagram $D$ for $wv^{-1}$ with a $w$-arc $A$, which may be assumed to be empty by Theorem 5.15. Using Propositions 5.17, 5.19 and 5.20, we see that the shortcut $\alpha_A$ for $w_A$ induces a word $u$ that is shorter than $w$ and $k$-fellow travels with $w$ as needed. □

The above results lead to the following:

**Theorem 5.24.** *Let $\langle \Sigma | H \sqcup S\rangle$ be a prismatic presentation and, for a relation $h \in H$, let $|h|$ be its length as a word over $\Sigma \cup \Sigma^{-1}$. If $H$ is empty, or if there is a constant*

$$n = \sup\{|h| : h \in H\} < \infty,$$

*then the Cayley graph of the group given by the prismatic presentation with respect to $\Sigma$ has the falsification by fellow traveller property. In particular, if the generating set $\Sigma$ and the set of relations $H$ are finite then the group has the falsification by fellow traveller property with respect to the standard generating set.*

Groups with the falsification by fellow traveller property have a decidable word problem, that is, we can tell whether a word over the generators represents the trivial element of the group or not. In [3, §2.3], the authors propose an algorithm for solving the word problem in trickle groups that resembles Tits algorithm for Coxeter groups [18]. As a consequence of the geodesic characterisation for prismatic presentations given in Theorem 5.21, we also obtain a Tits-style algorithm that solves the word problem in groups with a prismatic presentation.

Tits algorithm for Coxeter groups, as well as its generalisation to trickle groups, consists of a rewriting system defined by two types of rules: the rules in the first family, called $M_I$, reduce the length of a word; the rules in the second family, $M_{II}$, do not change the length of a word and are reversible. In this setting, a word is $M$-reduced if it cannot be shortened by a sequence of applications of rules in $M_I \cup M_{II}$. It is important to note that, in the case of trickle groups, the generating set used is different from the standard one and, consequently, so is the definition of "shortening": if $x$ is a (standard) generator of order $n$ in the trickle group, all the words $x, x^2, \dots, x^{n-1}$ are also considered generators in this setting. The power of this rewriting system relies on the following facts:

(1) A word over the generators of a Coxeter [trickle] group is geodesic if and only if it is $M$-reduced.
(2) Two geodesic words over the generators of a Coxeter [trickle] group represent the same element if and only if one can be transformed into the other by a sequence of $M_{II}$ rules.

For prismatic presentations, we can define a rewriting system $M = M_I \cup M_{II}$ as above and obtain a statement equivalent to (1). It will also be clear what the obstruction to obtaining (2) is.

**LAM:** ojo, yo creo qie si las cabezas son impares la segunda parte también podemos deducirla. a lo mejor podríamos hablar, a futuro, de qué podemos decir con las cabezas pares... argumento tipo paridad como el que funcionó para automaticidad?

**Definition 5.25** (Tits-like rewriting system)**.** Let $\langle \Sigma \mid H \cup S\rangle$ be a prismatic presentation. We define the following rewriting system for the words over $\Sigma$. Letting $\varepsilon$ denote the empty word, set $M = M_I \cup M_{II}$, where

$$\begin{aligned} M_I = &\{u \to v \mid |v| < |u|,\ uv^{-1} \in H \text{ (up to cyclic permutation/inverse)}\} \cup \\ &\{x^\tau x^{-\tau} \to \varepsilon \mid x \in \Sigma, \tau \in \{1, -1\}\}, \\ M_{II} = &\{u \to v \mid |v| = |u|,\ uv^{-1} \in S \text{ (up to cyclic permutation/inverse)}\}. \end{aligned}$$

For $X \in \{M_I, M_{II}, M\}$, we write $u \xrightarrow{X} v$ if a word $v$ is obtained from a word $u$ in one $X$-rewriting step, and $u \xrightarrow{X} v$ if $v$ is obtained from $u$ in zero or more $X$-rewriting steps. We say that a word is $M$-reduced if it cannot be shortened by performing $M$-rewriting steps.

As anticipated, the rules in $M_I$ reduce the length of a word and are irreversible, whereas the rules in $M_{II}$ maintain the length of a word and are reversible – if $u \xrightarrow{M_{II}} v$, also $v \xrightarrow{M_{II}} u$. We have:

**Proposition 5.26.** *Let $\langle \Sigma \mid H \cup S \rangle$ be a prismatic presentation and $M = M_I \cup M_{II}$ as above. A word $w$ over $\Sigma$ is geodesic if and only if it is $M$-reduced.*

*Proof.* If a word $w$ is not $M$-reduced, it is clearly not geodesic with respect to the generating set $\Sigma$.

If $w$ is not geodesic, there is a shorter word $v$ representing the same group element. Take $D$ to be disc diagram for $wv^{-1}$ that is free of pathologies and siamese spiders, such diagram exists by Corollary 4.28. Such a diagram contains a $w$-arc by Theorem 5.21, and this $w$-arc can be emptied following Theorem 5.15. A diagram $D'$ for $wv^{-1}$ with an empty $w$-arc $A$ naturally induces a rewriting $w \xrightarrow{M*} w'$ that shows that $w$ is not $M$-reduced. This is because, as defined in Definitions 5.16 and 5.18, the shortcut $\alpha_A$ can be obtained from the word $w_A \subset \partial_w D'$. More precisely,

$$w_A \xrightarrow{M_{II}*} w'_A \xrightarrow{M_I} \alpha_A,$$

where there are as many $M_{II}$-moves applied as squares in the diagram enclosed by $w_A$ and $\alpha_A$ in $D'$, and the final $M_I$-move is of type $x^\tau x^{-\tau} \to \varepsilon$ if the $w$-arc $A$ is given by a corridor, and of type $u \to x$ for $ux^{-1} \in H$ if the $w$-arc $A$ is given by a spider. This rewriting sequence becomes clear by keeping in mind Figs. 23 and 24. ☐

Unlike in the case of Coxeter groups or more generally trickle groups, we cannot ensure that any two geodesic words are separated only by $M_{II}$-moves. If a prismatic presentation has a head relation of even length, take words $u$, $v$ of the same length such that $uv^{-1}$ equals an even head relation. Both $u$ and $v$ are geodesics but, unless $u = v$, one cannot rewrite one into the other by using $M_{II}$-moves (quasicommutations). The reason why this obstruction does not pose a problem in the equivalent result for trickle groups in [3, §2.4] is because the authors use a generating set of *syllables* when they define the rewriting system: heads in trickle groups are of the form $x^n = 1$, and they include all the words $x, x^2, \ldots, x^{n-1}$ as generators. In the example above, this means that the words $u$ and $v$ would be of length one and equal to the generator $x^{\frac{n}{2}}$, so technically $u$ can be transformed into $v$ with zero $M_{II}$ rules and (2) is trivially true.

However, the problem described above only arises due to heads of even length. The following result is the version of (2) that we are left with when we avoid even heads:

**Proposition 5.27.** *Let $\langle \Sigma \mid H \cup S \rangle$ be a prismatic presentation and $M = M_I \cup M_{II}$ as above. If no relations in $H$ have even length, two geodesic words $u$, $v$ over $\Sigma$ represent the same group element if and only if $u \xrightarrow{M_{II}*} v$.*

**LAM:** Paloma no tiene claro que merezca la pena decir esto, a discreción de Yago

**LAM:** Esto me lo estoy inventando sobre la marcha, a lo mejor es mentira o a lo mejor hay que mirarlo bien, pero debería ser cierto si lo es para RAAGs.

*Proof.* One direction ($\Leftarrow$) is clear. For the other ($\Rightarrow$), assume that $u$ and $v$ are geodesic words over $\Sigma$ and take $D$ to be a disc diagram for $uv^{-1}$ that is free of pathologies and siamese spiders, which can be done by Corollary 4.28. Such a diagram cannot contain any spiders, because the fact that all heads have odd length would ensure the existence of a $u$-arc or a $v$-arc and this would contradict the fact that these words are geodesic words (Theorem 5.21). The fact that $D$ is a squared diagram ensures that $u$ and $v$ are only separated by quasicommutations, that is, $M_{II}$-moves. □

## 6. Automaticity of groups with a prismatic presentation

The goal of this section is to prove that groups admitting a prismatic presentation are shortlex automatic, which we define next following [7, Chapters 2.3 and 2.5]. Recall the definition of $\mathrm{SL}_G$ from Notation 2.6 and, for a definition of a regular language, we refer the reader to [7, Chapter 1].

**Definition 6.1.** A group $G = \langle S \rangle$ is *shortlex automatic* if $\mathrm{SL}_G$ is a regular language and there exists a constant $K$ such that any two words $w, v \in \mathrm{SL}_G$ with $|w^{-1}v|_G \leq 1$ $K$-fellow travel.

From the previous section, we have that $\mathrm{SL}_G$ is regular for any group $G$ admitting a prismatic presentation. Now, to show that these groups are shortlex automatic, we need to see that there exists a constant $K$ such that any two words $w, v \in \mathrm{SL}_G$ with $|w^{-1}v|_G \leq 1$ $K$-fellow travel. Using the triangle inequality and the fact that $w$, $v$ are geodesic words, one can see that $|w| \in \{|v|-1, |v|, |v|+1\}$: the second case is studied in Proposition 6.2, the first and third case are symmetric and we study them in Proposition 6.3.Without loss of generality, we may assume that $w$ and $v$ have no common prefix – if they had a common prefix $z$, the words $w = zw'$and $v = zv'$ would $K$-fellow travel if and only if the words $w'$ and $v'$ $K$-fellow travelled.

**Proposition 6.2.** *Let $G$ be a group with prismatic presentation $\langle \Sigma | H \sqcup S \rangle$ and fix an ordering on $\Sigma \cup \Sigma^{-1}$. There exists a constant $K$ such that for every $w, u \in \mathrm{SL}_G$ and $x \in \Sigma \cup \Sigma^{-1}$ with $wx =_G u$ and $|u| = |w|$, $w$ and $u$ $K$-fellow travel.*

*Proof.* Let $w, u$ and $x$ be as in the statement and assume that they do not have a common prefix. Let $D$ be a disc diagram for $wxu^{-1}$ with no spider or corridor pathologies and no siamese spiders, such diagram exists by Corollary 4.28. We will denote by $C_x$ the corridor induced by $x$ in $D$.

Suppose that $C_x$ is not a leg of a spider. By symmetry, we may assume that the other extreme of $C_x$ is in $\partial_w D$. Because $w$ is geodesic, Theorem 5.21 implies that there are no $w$-arcs in the region enclosed by $C_x$ and $\partial_{wx} D$, so $C_x$ is an outermost $wx$-arc. Thus we may apply Theorem 5.15 to obtain a diagram as shown in Fig. 25, where the word $wx$ can be decomposed as $vew'x$ and $ew'x$ is the boundary of an empty $w$-arc induced by the edges $e$ and $x$. The concatenation of $v$ with the shortcut for this empty ($wx$)-arc gives us a path $\gamma$ (in blue) with the same endpoints as $u$ and satisfying that $|\gamma| < |w| = |u|$, in contradiction with $u$ being geodesic.

We may then suppose that the corridor $C_x$ in $D$ is a leg of a spider $S$. If $S$ has an odd number of legs, half of the legs not induced by $x$ end in $\partial_w D$ and the other half in $\partial_u D$ (if not, there would be a $w$-arc or a $u$-arc, which gives a contradiction by Theorem 5.21), so $S$ gives an outermost $wx$-arc. If $S$ has an even number of legs, half of them end in $\partial_w D$ (or $\partial_u D$) and half minus one of them end in $\partial_u D$ (or

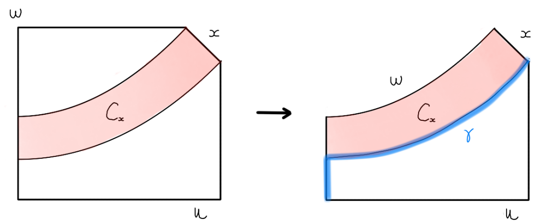



FIG. 25. $C_x$ is not a leg of a spider in Proposition 6.2.

$\partial_w D$), for the same reason as before. Without loss of generality, we assume that, if $S$ has an even number of legs, half of them end in $\partial_w D$, so $S$ gives an outermost $wx$-arc as well.

Applying Lemma 5.9, we may assume that the head of $S$ intersects $\partial D$ at the edge $x$, that the spider still gives an outermost $wx$-arc (call it $A_1$), that $I_{A_1}$ contains no spider heads and that $I_{A_1} \cup A_1$ contains no pathologies or spider pathologies. Now, applying Proposition 5.13, we can assume that $A_1$ is empty in $D$ (see the central diagrams in Fig. 26 and Fig. 27).

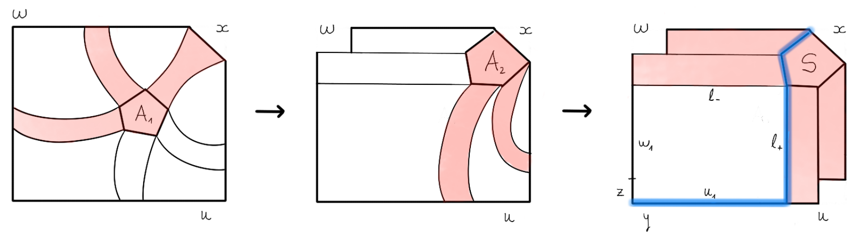



FIG. 26. $C_x$ is a leg of an odd-legged spider in Proposition 6.2.

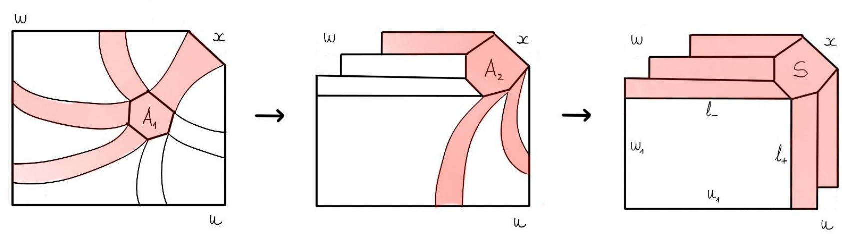



FIG. 27. $C_x$ is a leg of an even-legged spider in Proposition 6.2.

In the odd case, it is clear that the spider induces an outermost $xu^{-1}$-arc. For the even case, let us denote by $w'$ the suffix of $w$ whose first edge is the end of the leg ending in $\partial_w D$ which is adjacent to the leg induced by $x$. In this case the spider gives an outermost $w'xu^{-1}$-arc. Let us call the arc $A_2$ in both cases. Arguing as we did for $A_1$, we may assume that $A_2$ is empty in $D$ (see the right-most diagrams in Fig. 26 and Fig. 27).

Let us write $w = w_1w_2$ and $u = u_1u_2$, where $w_1$ and $u_1$ are the maximal prefixes of $w$ and $u$, respectively, that are part of $\partial(D - S)$. Also, let $y$ be the first letter of $u$ and let $z$ be the first letter of $w$. Moreover, we denote by $l_-$ and $l_+$ the sides of the legs of $S$ that are part of $\partial(D - S)$ and have a common vertex with $w_1$ and $u_1$, respectively. Applying Proposition 5.19 to the empty arc $A_2$, we have that the number of edges in $l_+$ (in $l_-$) is equal to the number of edges in $u_2$ (in $w_2$) which are not ends of legs of the spider $S$.

If $S$ has an odd number of legs, by symmetry we may assume that $y < z$. Then, it is clear that reading $u_1$, followed by $l_+$ and, from here, the shortest path in $\partial H$ with the same endpoint as $w$, yields a path of the same length as $w$ but lexicographically smaller, in contradiction with $w \in \mathrm{SL}_G$.

If $S$ has an even number of legs (say, $2r$). By Theorem 5.21 the region enclosed by $u_1l_+$ and $w_1l_-$ cannot contain any spiders with an odd number of legs (this would be in contradiction with $w$ and $u$ being geodesics). Therefore, all of the cells tessellating this region have an even number of edges and, since gluing diagrams of even perimeter gives diagrams of even perimeter, this region has even perimeter. However, applying Proposition 5.19, we see that $|u_1l_+| = |u| - (r-1)$ and $|w_1l_-| = |w| - r$, so the perimeter of the region enclosed by $u_1l_+$ and $w_1l_-$ is an odd number. This is a contradiction.

Since we reach a contradiction in every possible case, the statement is vacuously true. $\square$

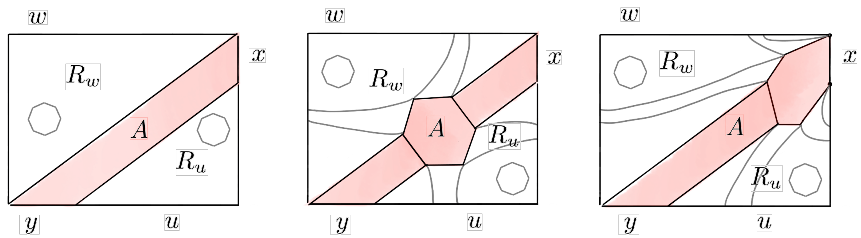
**Proposition 6.3.** *Let $G$ be a group with prismatic presentation $\langle \Sigma | H \sqcup S\rangle$ and fix an ordering on $\Sigma \cup \Sigma^{-1}$. There exists a constant $K$ such that for every $w, u \in \mathrm{SL}_G$ and $x \in \Sigma \cup \Sigma^{-1}$ with $wx =_G u$ and $|u| = |w| \pm 1$, $w$ and $u$ $K$-fellow travel.*

*Proof.* We shall give a proof for $|u| = |wx| = |w| + 1$, and the case $|u| = |w| - 1$ will follow from the symmetry between $w$ and $u$ in these two cases. Without loss of generality, we may assume that $w$ and $u$ have no common prefix.

Suppose then that $|u| = |wx| = |w| + 1$. If the word $wx$ belongs to $\mathrm{SL}_G$, we conclude that $u = wx$ and therefore $u$ and $w$ 1-fellow-travel. If the word $wx \notin \mathrm{SL}_G$, take $y$ to be the first letter of $u$. This letter is lexicographically smaller than the first letter of $w$ because $w$ and $u$ do not have a common prefix and because $u <_{\mathrm{lex}} wx$.

Let $D$ be a disc diagram for $wxu^{-1}$ chosen to be free of corridor and spider pathologies and with no siamese spiders as in Corollary 4.28. We will define a subcomplex $A$ inside $D$ that is an arc or a spider arc running from $y$ to $x$, we will then show that we can empty the region between $A$ and $\partial_w D$ and the region between $A$ and $\partial_u D$ and, similar to before, this will give us the fellow travelling property that we need. The definition of $A$ is slightly different depending on whether or not the corridor induced by $y$ is the leg of a spider, but the philosophy behind both definitions is the same.

If the corridor induced by $y$ determines a leg of a spider with $n$ legs, notice that $n$ must be even. If $n$ was odd, this spider would induce either a $u$-arc or a $wx$-arc but this cannot be the case since both $u$ and $wx$ are geodesic words from Theorem 5.21 ($u \in \mathrm{SL}_G$ and $|u| = |wx|$) . Thus, $n$ must be even. Moreover, since $D$ has no $u$-arcs, at least $\frac{n}{2}$ of these legs end in $wx$. If no leg ended in $x$, the spider would induce a $(y^{-1}w)$-arc $B$ which would be outermost (since $w$ is geodesic, there are no inner $w$-arcs) that we could empty using Theorem 5.15. Following the notation from Definition 5.18, the shortcut $\alpha_B$ for the word $(y^{-1}w)_B$ would induce a word




(A) $A$ is a corridor. (B) $A$ is part of a spider (C) Head pushed to $x$.

FIG. 28. Definition of the subcomplex $A$ and the regions $R_w$, $R_u$ (which may contain spider heads) depending on whether the edge $y$ defines a corridor or a spider. If $A$ belongs to a spider, its head can be pushed agains the edge $x$.

$w'$ starting with the letter $y$ which would be equivalent to $w$ in the group, with $|w'| \leq |w|$ because $|\alpha_B| < |(y^{-1}w)_B|$ and therefore with $w' <_{\text{lex}} w$. This would contradict the fact that $w \in \text{SL}_G$; thus, this spider has a leg ending in $x$, a leg ending in $y$ and leaving exactly $\frac{n}{2} - 1$ legs ending in $\partial_w D$ and $\frac{n}{2} - 1$ legs ending in $\partial_u D$. As shown in Fig. 28(B), define the subcomplex $A$ to be the arc given by the head of the spider, the leg induced by $y$ and the leg induced by $x$.

If $y$ determines a corridor whose other end is in $\partial D$, let $A$ be the complex given by this corridor: we will see that $A$ necessarily ends in $x$. $A$ cannot end in $u$, because then it would be a $w$-arc contradicting the fact that $u$ is a geodesic ($u \in \text{SL}_G$). Similar to before, if $A$ ends in $w$ it yields a $(y^{-1}w)$-arc that can be emptied and from whose shortcut a word $w'$ contradicting the fact that $w \in \text{SL}_G$ can be built. We conclude that the corridor $A$ starting at $y$ must have its other end in the edge $x$, as in Fig. 28(A).

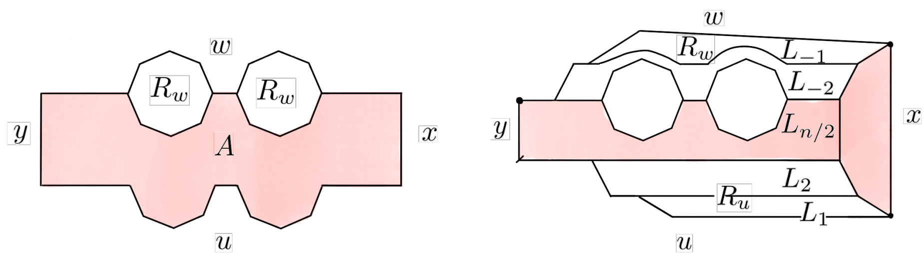
We now have a subcomplex $A$, which is both a $(y^{-1}wx)$-arc and a $(yux^{-1})$-arc. Up to a prismatic move we can assume that, if $A$ is given by a spider, its head is pushed against $x$ (see Fig. 28(C)). Now consider the region $R_w$ enclosed between $A$ and $\partial_w D$ and the region $R_u$ enclosed by $A$ and $\partial_u D$, yielding the partition $D = R_w \sqcup A \sqcup R_u$. Since $R_w \cap \partial_w D$ and $R_u \cap \partial_u D$ are labelled by geodesics, the region $R_w$ has no $w$-arcs and the region $R_u$ has no $u$-arcs. These regions have no corridor pathologies, spider pathologies or siamese spiders. We can retesselate $D$ to make $A$ into an empty $(y^{-1}wx)$-arc, and then retesselate the resulting diagram to make the corresponding $(yux^{-1})$-arc into an empty one as described in Theorem 5.15.

**PLL:** Ojo con este prismatic move, ¿cómo justificamos que no se crean patologías cpn el framework actual?**LAM:** juraria que esto ya lo hemos hablado, lo dejo por si acaso

If $A$ is given by a corridor, the retessellated diagram $D$ (strictly, $D'$, but we will keep the notation from $D$ for simplicity) will be given by $A \cup R_w$, where $R_w$ is now a union of spider heads that may have been pushed from $R_u$ before emptying the $(yux^{-1})$-arc as depicted in Fig. 29(A).

Following Notation 5.11: if $A$ is given by a spider $S$ the $(y^{-1}wx)$-arc involves the legs $L_{-\frac{n}{2}}, \ldots, L_0$ and the $(yux^{-1})$-arc involves the legs $L_0, \ldots, L_{\frac{n}{2}}$ of the spider $S$. After emptying both arcs as described above, the retessellated diagram $D$ will be equal to $S \sqcup (R_{-\frac{n}{2}-1} \cap R_{\frac{n}{2}+1})$, where $(R_{-\frac{n}{2}-1} \cap R_{\frac{n}{2}+1}) \subset R_w$ is a union of spider heads that may have been pushed from $R_u$. The resulting diagram is shown in Fig. 29(B).

Regardless of whether or not $A$ is given by a spider, the prismatic moves used in Lemma 5.9 to push heads, together with the absence of siamese spiders in the




(A) (Almost) empty corridor. (B) (Almost) empty spider.

FIG. 29. Once the $(yux^{-1})$-arc and the $(y^{-1}wx)$-arc defined by $A$ are emptied, all the heads from $R_u \cup R_w$ are pushed against $R_w$ and they all intersect $A$, so $w$ and $u$ $K$-fellow travel.

original diagram, ensures that each of the spider heads left in $R_w$ after emptying the $(yux^{-1})$-arc is attached to $A$ through at least one edge: in particular, the heads in $R_w$ cannot pile up and the distance between $w$ and $u$ is uniformly bounded as we detail below.

Let $m$ be the maximal order of a torsion generator in the trickle group or let $m = 0$ if there are no torsion generators: if $A$ is given by a corridor then $w$ and $u$ $k$-fellow travel asynchronously with

$$k = 1 + \left\lfloor \frac{m}{2} \right\rfloor,$$

where the addend 1 is the width of $A$ and the second addend corresponds to crossing the polygons in $R_w$; if $A$ is given by a spider of $n$ legs, $w$ and $u$ $k$-fellow travel asynchronously with

$$k = \frac{n-2}{2} + 1 + \left\lfloor \frac{m}{2} \right\rfloor + \frac{n-2}{2} = (n-1) + \left\lfloor \frac{m}{2} \right\rfloor,$$

where the two addends $\frac{n-2}{2}$ correspond respectively to the distance between $u$ and $A$ and to the distance between $w$ and the polygons in $R_w$ and the addend $1 + \left\lfloor \frac{m}{2} \right\rfloor$ has the same interpretation as in the case where $A$ is a corridor.

Given these observations, it is clear that one can choose a constant $K$ that works for arbitrary choices of words $w$ and $u$: it is enough to take

$$K = (m-1) + \left\lfloor \frac{m}{2} \right\rfloor, \quad \text{or} \quad K = 1$$

depending on whether the group has torsion generators or not.

□

**Theorem 6.4.** *Every group admitting a prismatic presentation with finite generating set is shortlex automatic.*

*Proof.* Let $G$ be a group with prismatic presentation $\langle \Sigma | H \sqcup S \rangle$, where we assume $\Sigma$ is finite. Fix any ordering of the elements of $\Sigma \cup \Sigma^{-1}$. As a consequence of Theorem 5.24, Lemma 2.8 and Proposition 2.9, we have that $\mathrm{SL}_G$ is a regular language and, by Proposition 6.2 and Proposition 6.3, there exists a constant $K$ such that any two words $w, v \in \mathrm{SL}_G$ with $|w^{-1}v|_G \leq 1$ $K$-fellow travel. □

It is natural to ask if the SL-automatic structure that we have provided (which is independent of the ordering that we choose for the generators) is also an SL-biautomatic structure. The following example shows that this is not always the case.

**Example 6.5.** Consider the right-angled Coxeter group given by this presentation

$$G = \langle x, y, z \mid [x, y] = 1, [y, z] = 1, x^2 = y^2 = z^2 = 1 \rangle.$$

If we have that $z < y < x$, the words $u = y^n x z^n$ and $w = z^n y^n$ belong to $\mathrm{SL}_G$ and satisfy that $xw =_G u$. Now, for any fellow-travelling constant $K$, $n$ can be chosen large enough so that $u$ and $w$ do not $K$-fellow travel.

## 7. Groups with a prismatic presentation: trickle and T-RAAGs

7.1. **Trickle groups.** The goal of this section is to show that the standard presentation of trickle groups is a prismatic presentation; that is, that it satisfies conditions **P1**-**P5** in Definition 3.6.

- Given a trickle graph $\Gamma$, the standard presentation of the trickle group $Tr(\Gamma)$ clearly satisfies condition **P1** (squares and heads) if we let $H$ be the set of relations of the form $x^{\mu(x)} = 1$ and let $S$ be the set of relations associated to quasi-commutations.
- As for condition **P2** (small pieces), observe that an edge in $\Gamma$ always joins different vertices $x \neq y$ and, therefore, any two adjacent edges in the 2-cell associated to the quasi-commutation relation $xy_x = yx_y$ between $x$ and $y$ must be labelled differently since $\varphi_x(y) = x \Leftrightarrow y = x$. This makes clear that no cyclic permutation of a relation in $S$ shares a subword of length 2 with a relation in $H$. Moreover, if we observe that two adjacent edges of a quasi-commutation 2-cell always determine the other two (as indicated in Fig. 30), we have that no cyclic permutation of a relation in $S$ shares a subword of length 2 with another relation in $S$. We conclude that Condition **P2** is satisfied.
- Condition **P5** (fusion of heads) is trivially satisfied because there is at most one head relation $x^{\mu(x)} = 1$ for each vertex $x$ in $\Gamma$, so two different relations in $H$ do not share any subword.

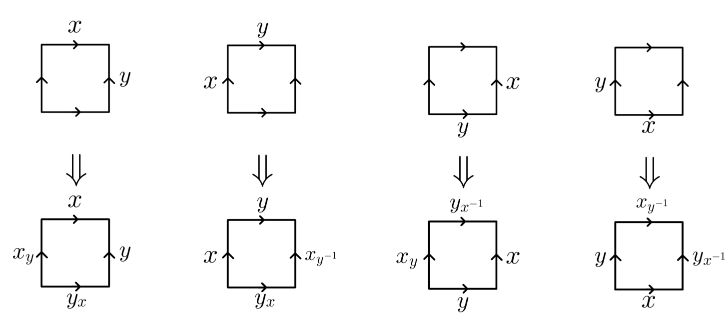



Fig. 30. Any two consecutive edges determine the rest.

In the process of writing this work, one of the authors in [3] brought to our attention that the cube completion in the Cayley complex of a trickle group can be deduced

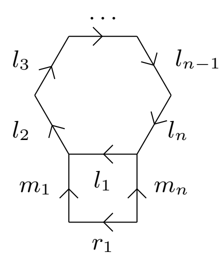



FIG. 31. Relations $\tau = l_1 \dots l_n$, $\sigma = l_1 m_1^{-1} r_1^{-1} m_n$ as in **P4**.

from the results in [3, Section 7] with regard to the sharp $\theta$-cube condition. For the readers not familiar with the ideas from Garside theory that their results require, we include a proof solely based on the definition of trickle groups.

The rest of this section is dedicated to showing that conditions **P3** and **P4** are satisfied by a trickle group presentation, we start by looking at condition **P4** (prism completion), for which we need these short lemmas.

**Lemma 7.1.** *Let $\Gamma$ be a trickle graph and let $x, y \in V(\Gamma)$. If $y < x$, then $y_x < x$.*

*Proof.* Notice that, if $y < x$, condition (a) ensures that $\{x, y\} \in E(\Gamma)$. By condition (d), we have that $x_x = x$. Now, if $y < x$, by condition (c) and the fact that $\varphi_x$ is a bijection on $\mathrm{Star}_\Gamma(x)$, we conclude that $y_x < x_x = x$. □

**Lemma 7.2.** *Let $\Gamma$ be a trickle graph, let $\{x, y\} \in E(\Gamma)$ and let $\varepsilon \in \{-1, 1\}$. Then $y_{x^\varepsilon} = y_{(x_y)^\varepsilon}$ and, as a consequence, $y_{(x_{y^i})^\varepsilon} = y_{x^\varepsilon}$ for any $i$ in $\mathbb{N}$.*

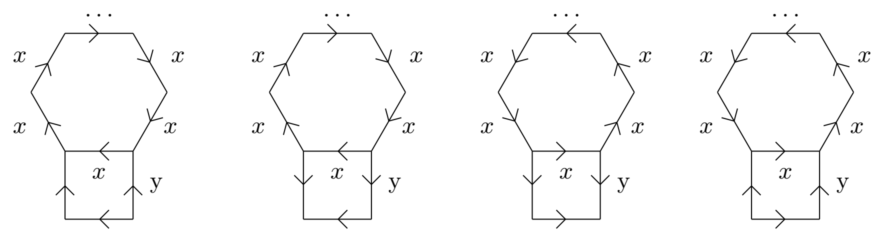
*Proof.* As $x \in \mathrm{Star}(y)$, if $x \not< y$, then by condition (d) $x_y = x$ and, therefore, $y_{x^\varepsilon} = y_{(x_y)^\varepsilon}$. If $x < y$, by Lemma 7.1, we also have $x_y < y$ and, applying condition (d), we obtain $y_{x^\varepsilon} = y = y_{(x_y)^\varepsilon}$. □

**Proposition 7.3.** *Let $\Gamma$ be a trickle graph and let $\langle \Sigma \mid H \cup S \rangle$ be the presentation for the corresponding trickle group, where the partition of relations $H \cup S$ is done as described above. This presentation satisfies condition **P4** (prism completion) from Definition 3.6.*

*Proof.* Consider relations $\tau = l_1 \dots l_n$ in $H$ and $\sigma = l_1 m_1^{-1} r_1^{-1} m_n$ in $S$ as described in condition **P4** and depicted in Fig. 31. Note that, for a trickle group, there is a generator $x$ in $\Sigma$ with $\mu(x) = n$ such that $l_i = x$ for all $i = 1, \dots, n$, or $l_i = x^{-1}$ for all $i = 1, \dots, n$. Depending on whether $m_n$ is labelled by a generator $y$ in $\Sigma$ or $y^{-1}$ in $\Sigma^{-1}$, each of these two cases gives rise to two possible diagrams: a total of four cases which are depicted in Fig. 32, where the directed edges are labelled by $x, y \in \Sigma$. We argue why **P4** is satisfied in the cases depicted by Fig. 32(A) and Fig. 32(B), and cases Fig. 32(C) and Fig. 32(D) follow symmetrically. First note that, taking into account that two adjacent edges in a square determine the whole square (Fig. 30), the unlabelled edges in Fig. 32(A) and Fig. 32(B) can be written as shown in Fig. 33(A) and Fig. 33(B) respectively. Notice that $y_{x^{n-1}} = y_{x^{-1}}$ since $\mu(x) = n$ and the order of $\varphi_x$ divides $\mu(x)$.

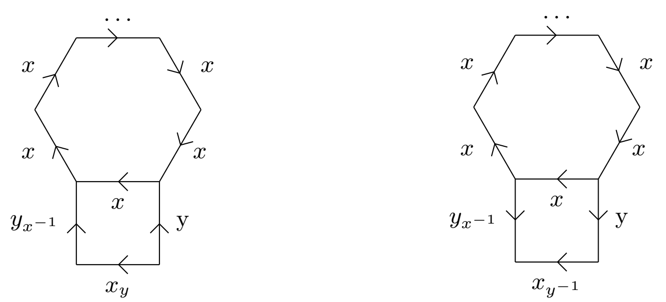
We first focus on the case shown in Fig. 33(A), where $\sigma_1$ is given by the square

$$\sigma_1 = x(y_{x^{n-1}})^{-1}(x_y)^{-1}y$$




(A) $l_1 = x$, $m_n = y$ (B) $l_1 = x$, $m_n = y^{-1}$ (C) $l_1 = x^{-1}, m_n = y$ (D) $l_1{=}x^{-1}, m_n{=}y^{-1}$

FIG. 32. The relations $\sigma$ and $\tau$ for **P4** in a trickle group, depending on whether $l_1$ and $m_n$ are labelled by a generator in $\Sigma$ or its inverse.




(A) $l_1 = x, m_n = y, r_1 = x_y, m_1 = y_{x^{-1}}$. (B) $l_1 = x$, $m_n = y^{-1}, r_1 = x_{y^{-1}}, m_1 = y^{-1}_{x^{-1}}$.

FIG. 33. The remaining labels for the edges of the square can be inferred from the labels in Fig. 32

corresponding to a quasi-commutation between the generator $x$ and the generator $y_{x^{n-1}}$. We claim that taking $r_i = x_y$ and $m_i = y_{x^{n-i}}$ for $i = 1, \dots, n$ yields the relations $\rho, \sigma_2, \dots, \sigma_n$ that fulfill condition **P4**. First, $\rho = r_1 \dots r_n = (x_y)^n$ clearly defines a relation in $H$ since $\mu(x) = \mu(x_y)$ by the definition of a trickle graph. Hence, we only need to check that

$$\sigma_i = l_i m_i^{-1} r_i^{-1} m_{i-1} = x(y_{x^{n-i}})^{-1}(x_y)^{-1} y_{x^{n-i+1}}$$

defines a relation in the trickle group, but this is the same as asking if

$$x_y y_{x^{n-i}} = y_{x^{n-i+1}} x.$$

Since $\sigma_1$ defines the relation $x_y y_{x^{n-1}} = y_x x$, we know that $\{x, y_{x^{n-1}}\} \in E(\Gamma)$ and therefore also $\{x, y_{x^{n-i}}\} \in E(\Gamma)$ for $i = 2, \dots, n$. The relation induced by this edge is, by definition,

$$x_{y_{x^{n-i}}} y_{x^{n-i}} = y_{x^{n-i}x} x,$$

which becomes the desired relation after noting that $x_{y_{x^{n-i}}} = x_y$ (Lemma 7.2).

Having shown that prism completion is possible in the case of Fig. 33(A), we proceed similarly for the case of Fig. 33(B). In this case, $\sigma_1$ is given by the square

$$\sigma_1 = x y_{x^{n-1}} (x_{y^{-1}})^{-1} y^{-1}$$

coming from the quasi-commutation relation between the generators $x_{y^{-1}}$ and $y_{x^{n-1}}$. We will see that taking $r_i = x_{y^{-1}}$ and $m_i = (y_{x^{n-i}})^{-1}$ for $i = 1, \dots, n$

yields the relations $\rho, \sigma_2, \dots, \sigma_n$ that fulfill condition **P4**. As before, $\rho = r_1 \dots r_n = (x_{y^{-1}})^n$ defines a relation in $H$ because $\mu(x) = \mu(x_{y^{-1}})$, so we only need to verify that

$$\sigma_i = l_i m_i^{-1} r_i^{-1} m_{i-1} = x y_{x^{n-i}} (x_{y^{-1}})^{-1} (y_{x^{n-i+1}})^{-1}$$

defines a square relation in the trickle group, equivalently, we need to see that

$$x y_{x^{n-i}} = y_{x^{n-i+1}} x_{y^{-1}}.$$

**LAM:** paloma sugería no dar explicación en la frase que empieza con 'But this is...' yo la he acortado por ahora, simplemente porque no es la quasicommutation "por definición" sino que hay que hacer una minicuenta para comprobarlo

But this is a quasi-commutation relation induced by $y_{x^{n-i}}$ and $x_{y^{-1}}$, which can be checked by taking into account that either $x_{y^k} = x$ or $y_{x^k} = y$ for all $k \in \mathbb{Z}$.

To ensure that this quasi-commutation is indeed a square relation in the trickle group, we need to check that $y_{x^{n-i}}$ and $x_{y^{-1}}$ are joined by an edge in $\Gamma$. This is a consequence of the fact that $x_{y^{-1}}$ and $y_{x^{-1}}$ share an edge in $\Gamma$, because this means that $(y_{x^{-1}})_{x^{n-i+1}}$ (which is equal to $y_{x^{n-i}}$) and $(x_{y^{-1}})_{x^{n-i+1}}$ share and edge in $\Gamma$, and $(x_{y^{-1}})_{x^{n-i+1}} = x_{y^{-1}}$ by Lemma 7.2. □

Proving that a trickle group presentation satisfies condition **P3** (cube completion) requires a bit more work and we need to take into consideration the possible orientations of the edges in the squares of a half-cube.

Because the square 2-cells in a trickle group are associated to relations of the type $y_x x = x_y y$, opposite edges in these squares have the same orientation. This means that the orientation of the edges in the boundary of the half-cube defined by three squares $\sigma_1$, $\sigma_2$ and $\sigma_3$ only depends on the orientation of the three edges that are incident to the unique vertex in the interior of the half-cube (each of these edges may be pointing towards or against this vertex). Thus, we may find four different types of half-cubes given by a trickle group presentation, which have been depicted in Figure 34.

Notice that these half-cubes come in pairs: the underlying directed hexagon is identical for half-cubes of types I and II, and the same happens for those of types III and IV. We will show that a half-cube of type I in $X_\Gamma$ admits a hexagonal move to a half-cube of type II, and vice versa; the same will be shown for types III and IV (as depicted in Figure 35). This, which is encompassed by Propositions 7.5, 7.8 and 7.9, proves that condition **P3** (cube completion) in Definition 3.6 is satisfied by trickle groups.

Proposition 7.5 states that a hexagonal move can be applied to a half-cube of type II to obtain a half-cube of type I, and that a hexagonal move can be applied to a half-cube of type IV to a half-cube of type III. In order to prove this statement, Lemma 7.4 is needed to make some observations on how different generators of the group relate to each other.

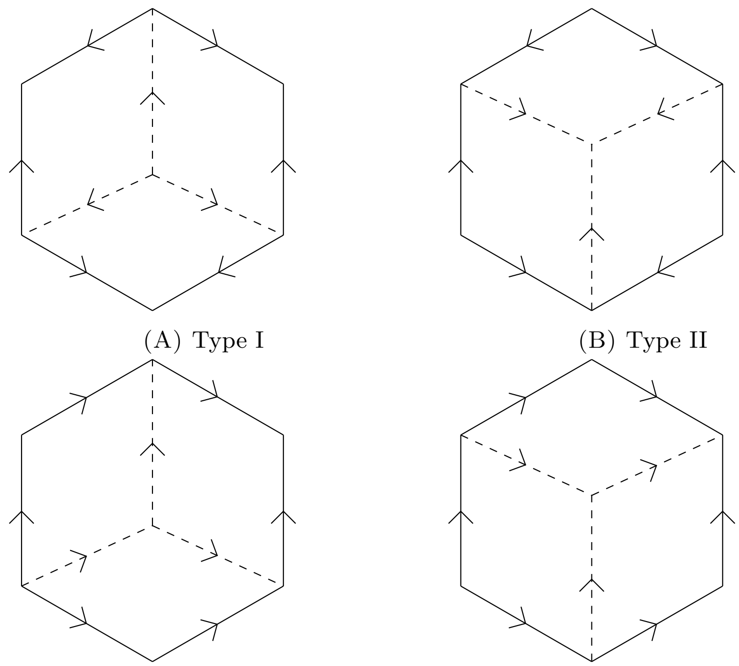
**Lemma 7.4.** *Let $\Gamma$ be a trickle graph and let $x, y, z \in V(\Gamma)$. If the vertices $x, y$ and $z$ are pairwise joined by edges in $\Gamma$, then the following equality holds*

$$z_{x y_x} = z_{y x_y}. \tag{1}$$

*Proof.* Considering the possible order relations between the vertices, we need to verify the equality in several cases.

Case 1: Assume $x < z$. By condition (c), we also have $x_y < z_y$. Then, by condition (d), $z_x = z$ and $z_{y x_y} = (z_y)_{x_y} = z_y$, so the equality that we need to verify is $z_{y_x} = z_y$.

(1.1) If $y < x$, using Lemma 7.1 and the transitivity of $\leq$, we have that $y < z$ and $y_x < z$, so $z_{y_x} = z = z_y$ by condition (d).



(A) Type I (B) Type II

(C) Type III (D) Type IV

FIG. 34. Four types of half-cubes over trickle groups.

(1.2) If $y \not< x$, condition (d) ensures that $y_x = y$ and we obtain $z_{y_x} = z_y$.

Case 2: Assume $y < z$. This case is symmetrical to the previous one, so $z_{xy_x} = z_{yx_y}$ holds.

Case 3: Assume $x \not< z$ and $y \not< z$.

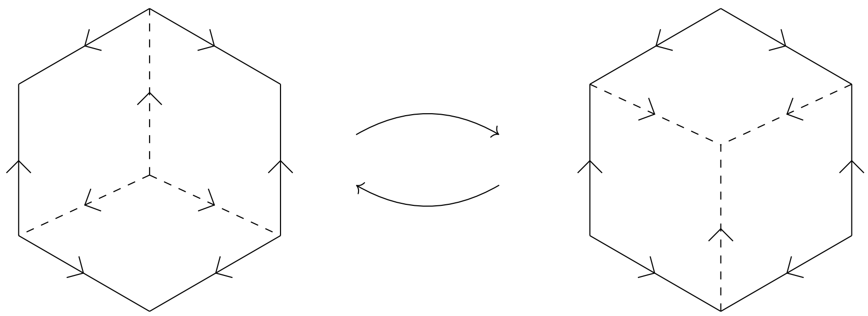
(3.1) If $x||y$, condition (d) yields $x_y = x$ and $y_x = y$, so the equality that we need to verify becomes $z_{xy} = z_{yx}$. Now, by condition (b), $z$ cannot be smaller than both $x$ and $y$, so it must be that either $x||z$ or $y||z$. Without loss of generality, suppose $y||z$. By condition (c), we have $y = y_x||z_x$ and, using condition (d), we obtain $z_{xy} = z_x = z_{yx}$.

(3.2) If $y < x$, there are two possibilities:

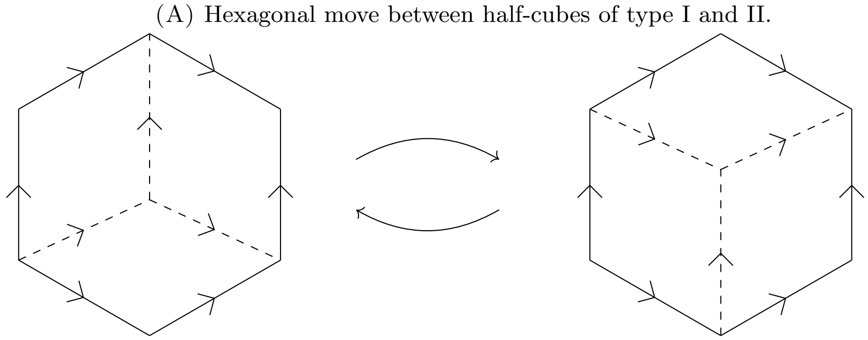
(3.2.1) If $z < y$, we have the chain $z < y < x$, so the equality $z_{yx} = z_{xy_x}$ is true by condition (g). Condition (d) gives $x_y = x$, so we have $z_{xy_x} = z_{yx} = z_{yx_y}$.

(3.2.2) If $z \not< y$, since $y \not< z$, we must have $z||y$. And, since $x \not< z$, there are two subcases:

- If $z < x$, we will see that $z_{xy_x} = z_x = z_{yx_y}$. Indeed, from $z||y$, by condition (c), we have that $z_x||y_x$, which implies that $z_{xy_x} = z_x$ by condition (d). On the other hand, condition (d) ensures $z_y = z$ and $x_y = x$, so we have $z_{yx_y} = z_x$.
- If $z||x$, we have that $z_{xy_x} = z_x = z_{yx_y}$. Indeed, condition (c) gives us that $z_x||y_x$, so $z_{xy_x} = z_x$ by condition (d). Now, because $z||y$ and $y < x$, condition (d) yields $z_{yx_y} = z_{x_y} = z_x$.



(A) Hexagonal move between half-cubes of type I and II.



(B) Hexagonal move between half-cubes of type III and IV.

FIG. 35. The four types of half-cubes come in complementary pairs.

(3.3) If $x < y$, the situation is analogous to the Case 3.2 due to the symmetry of the equation (1) with respect to $x$ and $y$.

□

The above lemmas are all we need to prove that the "right to left" retessellations in Fig. 35, are possible:

**Proposition 7.5** (II to I and IV to III)**.** *Let $\Gamma$ be a trickle graph.*

*(a) If $X_\Gamma$ contains a half-cube of type II, then it admits a hexagonal move that yields a half-cube of type I. And,*
*(b) if $X_\Gamma$ contains a half-cube of type IV, then it admits a hexagonal move that yields a half-cube of type III.*

*Proof.* (a) Consider the half-cube of type II and label the three incoming edges at the central vertex by $x, y, z \in V\Gamma$ as shown in Fig. 36(B). Since the three 2-cells come from quasi-commuting relations, their boundaries must be labelled by the relations

$$\begin{aligned} x_y y &= y_x x, \\ x_z z &= z_x x, \\ y_z z &= z_y y. \end{aligned}$$

By definition of the presentation, we have $\{x, y\}, \{x, z\}, \{y, z\} \in E\Gamma$. Firstly, this means that $x, y, z \in \mathrm{Star}(x) \cap \mathrm{Star}(y) \cup \mathrm{Star}(z)$ and, since $\varphi_x, \varphi_y, \varphi_z$ are graph automorphisms, we also have that $\{x_z, y_z\}, \{x_y, z_y\}, \{y_x, z_x\} \in E\Gamma$ and so the following relations hold:

$$\begin{aligned} x_{zy_z} y_z &= y_{zx_z} x_z, \\ x_{yz_y} z_y &= z_{yx_y} x_y, \\ y_{xz_x} z_x &= z_{xy_x} y_x. \end{aligned}$$

Secondly, we are under the hypotheses of Lemma 7.4, so we have the equalities $x_{zy_z} = x_{yz_y}$, $y_{zx_z} = y_{xz_x}$ and $z_{yx_y} = z_{xy_x}$. As a consequence, the 2-cells given by the relations above can be glued in order to retessellates the half-cube (see Fig. 36(A)).

(b) Consider the half-cube of type IV. Let $v$ be the right top vertex. As shown in Fig. 36(D), label the three incoming edges at $v$ by $x, y, z \in V\Gamma$, being $z$ the only edge not in the boundary. The two squares that have $v$ as one of its vertices come from quasi-commuting relations, so we have that $x, y \in \mathrm{Star}(z)$ and that the boundaries of the squares must be labelled by the relations

$$\begin{aligned} z_x x &= x_z z, \\ z_y y &= y_z z. \end{aligned}$$

The remaining square also comes from a quasi-commuting relation, so its associated relation must be of the form

$$x_{zy_z} y_z = y_{zx_z} x_z.$$

From this relation, we see that $\{x_z, y_z\} \in E\Gamma$ and both vertices belong to $\mathrm{Star}(z)$. As $\varphi_z$ is a graph automorphism, we obtain that $\{x, y\} \in E\Gamma$ and therefore $x, y, z$ are pairwise connected and the hypotheses of Lemma 7.4 hold. Because $\varphi_x$ and $\varphi_y$ are graph automorphisms, we get that $\varphi_x(\{y, z\}) = \{y_x, z_x\} \in E\Gamma$ and $\varphi_y(\{x, z\}) = \{x_y, z_y\} \in E\Gamma$. As we also have $\{x, y\} \in E\Gamma$, we obtain the following relations from the presentation:

$$\begin{aligned} x_y y &= y_x x, \\ y_{xz_x} z_x &= z_{xy_x} y_x, \\ x_{yz_y} z_y &= z_{yx_y} x_y. \end{aligned}$$

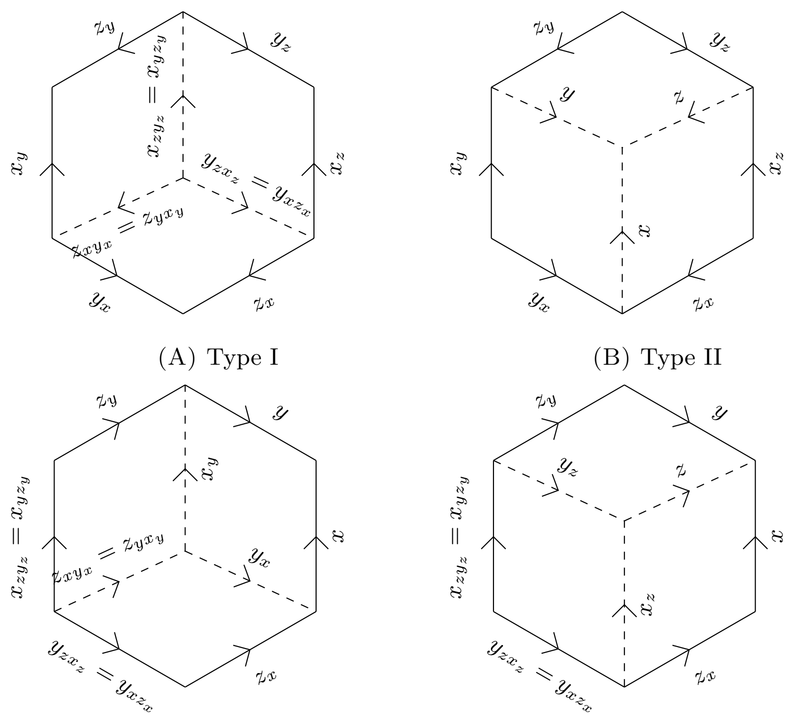
Now we can use these new relations to retessellate the half-cube using the equalities derived from Lemma 7.4, $x_{zy_z} = x_{yz_y}$, $y_{zx_z} = y_{xz_x}$ and $z_{yx_y} = z_{xy_x}$ (see Fig. 36(C)).

□

Propositions 7.8 and 7.9 prove that the remaining hexagonal moves (the "left-to-right" ones in Fig. 35) are also possible in trickle groups by showing that these are just inverse retessellations of the previous ones. Similar to before, in order to prove these statements Lemmas 7.6 and 7.7 make some preliminary observations relating the generators in the trickle group.

**Lemma 7.6.** *Let $\Gamma$ be a trickle graph and let $\{x, y\} \in E(\Gamma)$. Then, $x_y < y_x$ if and only if $x < y$.*

*Proof.* ($\Rightarrow$) Suppose that $x \not< y$. Then, by condition (d), $x_y = x$ and we have that $x_x = x = x_y < y_x$ and, by condition (c), it must be that $x < y$, a contradiction. ($\Leftarrow$) If $x < y$, by condition (d), $y_x = y$. By Lemma 7.1, $x_y < y = y_x$. □



(A) Type I (B) Type II

(C) Type III (D) Type IV

FIG. 36. Tesselated hexagons can always be labelled like this for some generators $x$, $y$, $z$.

**Lemma 7.7.** *Let $\Gamma$ be a trickle graph. If $\{x,y\}\in E\Gamma$, then $\{x,y_x\},\{x_y,y\},\{x_y,y_x\}\in E\Gamma$. In particular, the generators that label adjacent edges in a 2-cell of $X_\Gamma$ correspond to adjacent vertices in $\Gamma$.*

*Proof.* Since $\varphi_x$ and $\varphi_y$ are graph automorphisms and $\{x,y\}\in E\Gamma$, we have that $\{x,y_x\}=\varphi_x(\{x,y\})\in E\Gamma$ and $\{x_y,y\}=\varphi_y(\{x,y\})\in E\Gamma$. To conclude, just recall that we always have either $x_y=x$ or $y_x=y$, so $\{x_y,y_x\}=\{x,y_x\}\in E\Gamma$ or $\{x_y,y_x\}=\{x_y,y\}\in E\Gamma$.

□

**Proposition 7.8** (Type III to type IV)**.** *Let $\Gamma$ be a trickle graph. If the standard presentation of the corresponding trickle group yields a half-cube of type III, then there exist pairwise adjacent vertices $x,y,z\in V(\Gamma)$ such that the edges of the tessellated hexagon admit the labelling in Figure 36(C).*

*Proof.* A half-cube of type III can always be labelled as shown in Figure 37, where $x,y,p,q\in V(\Gamma)$ satisfy that $\{x,y\},\{p,x_y\},\{q,y_x\}\in E(\Gamma)$ and $p_{x_y}=q_{y_x}$. Now, to show that we can have the labelling in Figure 36(C), it is enough to find $z\in V(\Gamma)$ adjacent to both $x$ and $y$ satisfying that $z_y=p$ and $z_x=q$.

We know that $\{q,y_x\}\in E\Gamma$ so, by Lemma 7.7, $\{q_{y_x},y_x\}\in E\Gamma$. Since by hypothesis $p_{x_y}=q_{y_x}$ we have that $\{p_{x_y},y_x\}\in E\Gamma$ and now, by Lemma 7.2, we

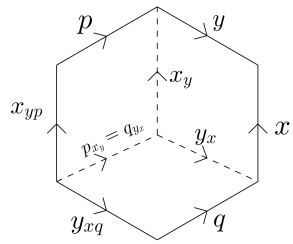



FIG. 37. Labelling for a type III hexagon.

see that $\{p_{x_y}, y_{x_y}\} \in E\Gamma$. Finally, since the automorphism $\varphi_{x_y}^{-1}$ preserves adjacency, we get that $\{p, y\} \in E\Gamma$. Since $\varphi_{x_y}$ preserves adjacency, we have that $\{p, y\} \in E(\Gamma)$. A symmetric argument gives that $\{q, x\} \in E(\Gamma)$ as well. Now we can take $z', z'' \in V(\Gamma)$ such that $z'_y = p$ and $z''_x = q$. Clearly, $z'$ is adjacent to $y$. Since $\{z'_y = p, x_y\} \in E(\Gamma)$ and $\varphi_y$ preserves adjacency, $z'$ is also adjacent to $x$. Thus, we may apply Lemma 7.4 to obtain $z'_{xy_x} = z'_{yx_y} = p_{x_y} = q_{y_x} = z''_{xy_x}$ and, because $\varphi_{y_x} \circ \varphi_x$ is injective, we conclude that $z' = z''$. Taking $z = z' = z''$ gives the desired labelling. □

**Proposition 7.9** (Type I to type II)**.** *Let $\Gamma$ be a trickle graph. If the standard presentation of the corresponding trickle group yields a half-cube of type I, then there exist pairwise adjacent vertices $x, y, z \in V(\Gamma)$ such that the edges of the tessellated hexagon admit the labelling in Figure 36(A).*

*Proof.* Given a half-cube of type I, there must be $p, q, r, s, u, v \in V(\Gamma)$ satisfying $u_p = v_r$, $p_u = q_s$ and $r_v = s_q$ so that edges can be labelled as in Figure 38. The goal now is to find pairwise adjacent vertices $x, y, z \in V(\Gamma)$ such that $u = x_y$, $v = x_z$, $r = y_z$, $s = y_x$, $p = z_y$ and $q = z_x$. We will do that at the very last stage of the proof.

We will begin showing that, without loss of generality, we can make some assumptions on the partial order restricted to $\{u_p, p_u, r_v\}$.

Notice that there are only five partial orders on a set of three elements $\{a, b, c\}$ up to order-isomorphism:

- the linear order: $a < b < c$,
- the order with a global minimum: $a < b, a < c$ and $b||c$.
- the order with a global maximum: $a||b, a < c$ and $b < c$.
- the order with a single chain of length 2: $b < c$, $a||b$ and $a||c$.
- the order where all elements are incomparable: $a||b, b||c$ and $a||c$.

On the five cases, it always hold that $a$ is smaller or incomparable to $b$ and $c$ and that $b$ is smaller or incomparable to $c$. By the symmetry of the configuration in Fig. 38, we can assume, without loss of generality, that the following three facts hold:

(1) $u_p$ is smaller than or incomparable to $r_v$,
(2) $u_p$ is smaller than or incomparable to $p_u$,
(3) $r_v$ is smaller than or incomparable to $p_u$.

Notice that $p_u = q_s$ is greater than or incomparable to the other elements. In particular, we have that $u_p < p_u$ or $u_p||p_u$ which, by Lemma 7.6, implies that $u < p$ or $u||p$ and, by condition (d), we have that $p_u = p$. Similarly, starting from

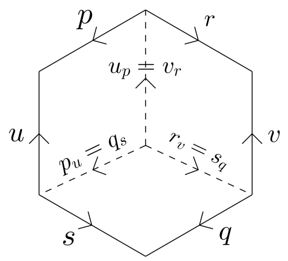



FIG. 38. Labelling for a half-cube of type I.

$r_v = s_q < q_s = p_u$ or $s_q||q_s$, we conclude that $q_s = q$. Therefore, we have that $p = p_u = q_s = q$. In particular,

$$p_s = p. \tag{2}$$

Observe as well that $u_p = v_r < r_v$ or $v_r||r_v$. Again, the previous argument gives us that $r = r_v = s_q$ and, as we have shown that $p = q$, we have that

$$s_p = r. \tag{3}$$

We claim that $\{u, s\} \in E\Gamma$ and $\{v, p\} \in E\Gamma$. Indeed, since they are labels of consecutive edges of a square 2-cell, $u_p$ and $s_q = s_p$ are adjacent in $\Gamma$ and since $\varphi_p$ is a graph automorphism, we obtain that $\{u, s\} \in E(\Gamma)$. Also, since $r = r_v < p_u$ or $r = r_v||p_u$ and $p_u = p$, we conclude that either $r < p$ or $r||p$ and therefore $p_r = p$. Given that $\{v_r, p_r = p\} \in E(\Gamma)$ and that adjacency is preserved by $\varphi_r$, we see that $\{v, p = z\} \in E(\Gamma)$.

By the previous paragraph, we can take $x' = \varphi_s^{-1}(u)$ and $x'' = \varphi_p^{-1}(v)$ and, using (2) and (3) respectively, we have

$$u_p = x'_{sp} = x'_{sp_s} \text{ and } v_r = x''_{pr} = x''_{ps_p}.$$

Now, since $u_p = v_r$, we have that $x'_{sp_s} = x''_{pr_p}$. Observe that $\{x'_s = u, p = p_s\} \in E\Gamma$ and, as $\varphi_s$ preserves adjacency, we have that $\{x', p\} \in E\Gamma$. As $x'_s \in \mathrm{Star}(s)$, $\{x', s\} \in E\Gamma$. Finally, as $p = q$, we see that $\{p, s\} \in E\Gamma$. Since we have checked that $p, s, x'$ are pairwise adjacent, we may use Lemma 7.4 and get that $x'_{sp_s} = x'_{ps_p}$. Therefore, as $u_p = v_r$, we have $x'_{ps_p} = x''_{ps_p}$ and now the injectivity of the vertex automorphisms gives us that $x' = x''$.

Set $x = x' = x''$, $y = s$ and $z = p$. Notice that by the previous paragraph, we know that $x, y, z$ are pairwise adjacent. We now proceed to verify that $u = x_y$, $v = x_z$, $r = y_z$, $s = y_x$, $p = z_y$ and $q = z_x$. First, we saw that $u = x'_s = x_y$. Second, we saw that $v = x''_p = x_z$. Third, by Eq. (3), we have that $r = y_z$. Fourth, because $u_q < s_q$ or $u_q||s_q$, using condition (c), we have that $u < s$ or $u||s$ and, using that $u = x_s$, $s = s_s$ and applying condition (c), we deduce that $x < s$ or $x||s$. So, by condition (d), $y_x = s_x = s$. Fifth, by Eq. (2), we have that $z_y = p$. Sixth, because $u_p < p_u$ or $u_p||p_u$, we have that $x_{sp} < p_{sp}$ or $x_{sp}||p_{sp}$ and condition (c) tells us that $x < p$ or $x||p$. And, by condition (d), $z_x = p_x = p = q$. □

As a consequence of Propositions 7.5, 7.8 and 7.9, we have:

**Proposition 7.10.** *The standard presentation of a trickle group $Tr(\Gamma)$ satisfies condition **P3** in Definition 3.6.*

Propositions 7.3 and 7.10, together with the observations made at the beginning of this section, allow us to conclude:

**Theorem 7.11.** *The standard presentation of a trickle group $Tr(\Gamma)$ is a prismatic presentation.*

7.2. **Twisted right-angled Artin groups.** This subsection is devoted to showing that the standard presentation for a T-RAAGs admit a prismatic presentation, which will provide examples different from trickle groups.

**Remark 7.12.** Twisted right-angled Artin groups are not trickle groups. In [9], the author shows that trickle groups where all the generators have infinite order must be orderable, but it is known by the results provided in [2] that there exist T-RAAGs which are not orderable.

Let $\Gamma$ be a mixed graph and let $\langle\sigma|R\rangle$ be the standard presentation of a T-RAAG that is determined by $\Gamma$. To see that this is a prismatic presentation, we show that **P1**-**P5** hold.

To show that **P1** holds, it is enough to take $H=\emptyset$ and $S=R$, since all relators are of the form $aba^{-1}b^{-1}$ or $abab^{-1}$, which are clearly cyclically reduced words of length four.

To check **P2**, observe that any two adjacent edges of a 2-cell coming from a relator in $R$ are labelled by different generators, so they determine the pair of edges in $\Gamma$ that gives rise to the relator associated to the 2-cell. Since each pair of vertices in $\Gamma$ may be joined by a unique edge (directed or undirected), it is impossible to find two different squares with two adjacent edges having the same labelling. Therefore, no cyclic permutation of a square relation shares a subword of length two with another relation.

Let us now verify **P3**. Suppose $\sigma_1, \sigma_2, \sigma_3$ are square relations that factorize as in **P3**. By looking at the generators that appear in the relation, we know which vertices $a$ and $b$ of $\Gamma$ give rise to this relation and, depending on the shape of the relation ($aba^{-1}b^{-1}$ or $abab^{-1}$), we know if $a$ and $b$ are joined by a directed or undirected edge and, in the directed case, whether the edge points at $a$ or at $b$. This determines a triangle in $\Gamma$ that can only be of one of the seven types indicated in Fig. 39.

For each of these triangles, we have cubes in the Cayley complex of the T-RAAG associated to $\Gamma$ as in Fig. 39. In order to obtain the square relations $\tau_1, \tau_2, \tau_3$ in **P3**, we simply take the relations associated to the squares in the complement of $\sigma_1, \sigma_2, \sigma_3$ in the corresponding cube. Thus, **P3** is satisfied.

As for conditions **P4** and **P5**, they vacuously hold because we have taken $H$ to be the empty subset. This discussion leads as to the following result:

**Theorem 7.13.** *Let $\Gamma$ be a mixed graph. The T-RAAG associated to $\Gamma$ admits a prismatic presentation.*

It is possible to find examples of prismatic presentations among torsion T-RAAGs, but restrictions must be put on the labelling of the vertices of the corresponding decorated mixed graph.

**Corollary 7.14.** *Let $\Gamma=(V,E,D,\iota,\tau,\mu)$ be a decorated mixed graph where the labelling of the vertices satisfies that, for every $[u,v\rangle$ $v\in D$, $\mu(v)$ is even or $\mu(v)=\infty$.*

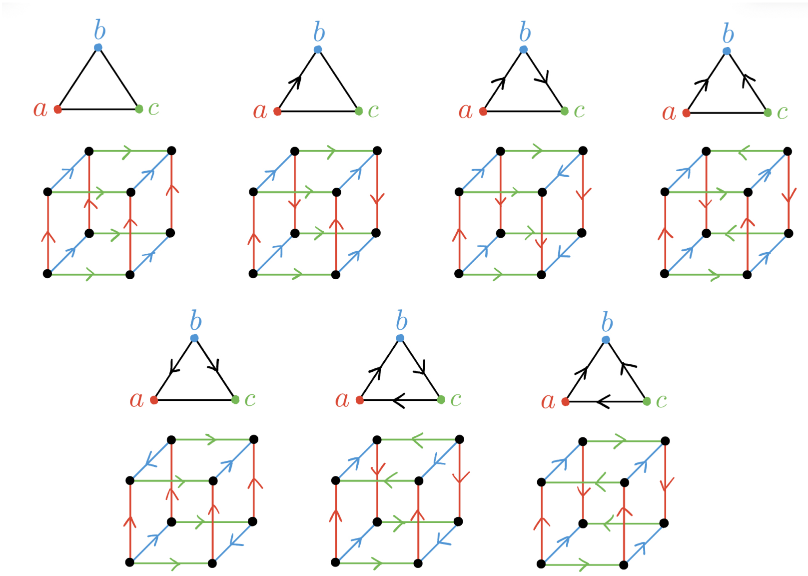



FIG. 39. Cubes in the Cayley complex of a T-RAAG.

*Proof.* To show that **P1** holds, it is enough to take $H = \{v^{\mu(v)} : v \in V\Gamma \text{ with } \mu(v) \neq \infty\}$ and $S$ as its complement in the set of relators of the presentation. Notice that $S$ is the same set of relators that makes the presentation of the underlying T-RAAG prismatic, so it is clear by the arguments above that **P3** is satisfied. Let us check the other conditions.

For **P2**, observe that consecutive edges of a head have the same label, which does not happen for any pair of consecutive edges in a square, so no head can be glued along two consecutive edges to a square. The same is true for two squares, by the same argument we used for T-RAAGs.

To show that **P4** holds, let $\tau$ be a head and let $\sigma_1$ be a square sharing an edge labelled by $v \in V\Gamma$ with $\tau$. Let $u \neq v$ the vertex that labels the other edges of $\sigma_1$. By definition $\{u, v\} \in E\Gamma$ and we have several possibilities.

- The edge is of type $[u, v]$. Then one can build a prism gluing $\mu(v)$ copies of $\sigma_1$ and two copies of $\tau$. Observe that the two bases of the prism will have parallel orientations.
- The edge is of type $[u, v\rangle$. Then one can build a prism gluing $\mu(v)$ copies of $\sigma_1$ and two copies of $\tau$. Notice that this is only possible because $\mu(v)$ is even (the edges joining the two bases of the prism present alternate orientations). In this case the two bases have parallel orientations as well.
- The edge is of type $[v, u\rangle$. Then one can build a prism gluing $\mu(v)$ copies of $\sigma_1$ and two copies of $\tau$. Observe that the two bases of the prism will have opposite orientations.

Finally, to verify **P5**, just notice that two heads that share an edge must be induced by the same relation so the cyclic reduction of the concatenation of the corresponding relators is the empty word.

□

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